

Factor Modelling and Benchmarking of Hedge Funds: Can passive investments in hedge fund strategies deliver?

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Abstract:

The hedge fund industry is starting to recognize that a main part of its returns corresponds to risk premia rather than market inefficiencies, i.e. from “beta” instead of “alpha”. This has some implication for the industry and investors, among which is the endeavor to construct investable benchmarks for hedge funds on the basis of an analysis of the underlying systematic risk factors and a subsequent replication of the corresponding risk premia with generic trading systems. The question touches further rationale on the sense and nonsense of the currently available investable versions of hedge fund indices. If possible, investable benchmarks based on risk factor analysis and replication offers a valid, theoretically more sound, and cheaper alternative to the currently offered hedge fund index products, especially as the latter reveal themselves more and more as questionable from a theoretical as well as practical standpoint. This article reflects on this most recent discussion within the global hedge fund industry about the “beta versus alpha” controversy, investable hedge fund indices, and finally, capacity issues. It illustrates how the current research activities in the quant groups of the large investment banks and financial academic centers might turn the hedge fund industry upside down in coming years. This article offers a follow up discussion on the broader treatment on the subject in the author’s book “Through the Alpha Smoke Screens: A Guide to Hedge Fund Return Sources”

Introduction

The debate on sources of hedge fund returns is one of the subjects creating the most heated discussion within the hedge fund industry. The industry appears to be split in two camps: Following results of substantial research, the proponents on the one side claim that the essential part of hedge fund returns come from the funds' exposure to systematic risks, i.e. comes from their betas. Conversely, the "alpha protagonists" argue that hedge fund returns depend mostly on the specific skill of the hedge fund managers, a claim that they express in characterising the hedge fund industry as an "absolute return" or "alpha generation" industry. As usual, the truth is likely to fall within the two extremes. Based on an increasing amount of empirical evidence, we can identify hedge fund returns as a (time-varying) mixture of both, systematic risk exposures (beta) and skill based absolute returns (alpha). However, the fundamental question is: How much is beta, and how much is alpha?

There is no consensus definition of 'alpha', and correspondingly there is no consensus model in the hedge fund industry for directly describing the alpha part of hedge fund returns. We define alpha as the part of the return that cannot be explained by the exposure to systematic risk factors in the global capital markets and it thus the return part that stems from the unique ability and skill set of the hedge fund manager. There is more agreement in modeling of their beta returns, i.e. the systematic risk exposures of hedge funds, which will give us a starting point for decomposition of hedge fund returns into 'alpha' and 'beta' components. We begin with stating the obvious: It is generally not easy to isolate the alpha from the beta in any active investment strategy. But for hedge funds it is not just difficult to separate the two, it is already quite troublesome to distinguish them. We are simply not in a position to give the precise breakdown yet. In other words, the current excitement about hedge funds has not yet been subject to the necessary amount of academic scrutiny. However, we argue that the better part of the confusion around hedge fund returns arises from the inability of conventional risk measures and theories to properly measure the diverse risk factors of hedge funds. This is why only recently progress in academic research has started to provide us with a better idea about the different systematic risk exposures of hedge funds and thus give us more precise insights into their return sources¹. Academic research and investors alike begin to realize that that the "search of alpha" must begin with the "understanding of beta", the latter constituting an important – if not the most important - source of hedge fund returns.²

However, at the same time we are starting to realize that hedge fund beta is different from traditional beta. While both are the result of exposures to systematic risks in the global capital markets hedge fund beta is more complex than traditional beta. Some investors can live with a rather simple but illustrative scheme suggested by C. Asness³: If the specific return is available only to a handful investors and the scheme of extracting it cannot be simply specified by a systematic process, then it is most

¹ See the recently published book by Jaeger (2005) and references therein.

² Martin (2004) makes the pertinent point that measures of alpha inextricably depend on the definition of benchmarks or beta components, going on to identify ways in which techniques for measuring 'alpha' in a traditional asset management environment are inappropriate or otherwise undermined by the specific characteristics of hedge fund exposures. Moreover, most techniques for measuring hedge fund alpha tend to reward fund managers for model and benchmark misspecification, as imperfect specification of benchmark or 'beta' exposure tends to inflate alpha.

³ Asness (2004).

likely real alpha. If it can be specified in a systematic way, but it involves non-conventional techniques such as short selling, leverage and the use of derivatives (techniques which are often used to specifically characterize hedge funds), then it is possibly beta, however in an alternative form, which we will refer to as “alternative beta”. In the hedge fund industry “Alternative beta” is often sold as alpha, but is not real alpha as we define it.. If finally extracting the returns does not require any of these special “hedge fund techniques” but rather “long only investing”, then it is “traditional beta”.

But how do we model hedge fund returns explicitly and break them down into alpha, alternative beta and traditional beta? Ultimately, what we are looking for is a general equilibrium model, which relates hedge fund returns to their systematic risk exposures represented by directly observable market prices in the financial markets, similar to the Capital Asset Pricing Model for the equity markets⁴. This model does not exist yet in its entirety, but there exists today a growing amount of academic literature on systematic risk factors and hedge funds’ exposure to them (i.e. their factor loadings), including a variety of “alternative beta factors”. We acknowledge that the quality of the offered model differs strongly for the different hedge fund strategy sectors. In other words, there is a variable degree of explanatory power for (the variation of) hedge fund returns that factor models can offer across different strategy sectors. While Long/Short Equity has been well modeled in academic research⁵, models for some other strategies like Arbitrage strategies (Equity Market Neutral, Convertible Arbitrage) display rather limited explanatory power (i.e. low R-squared values).

This article aims to give reference to this academic effort and provide a coherent discussion on the current status of “beta versus alpha” controversy in the hedge fund industry. Literature references are given extensively. However, it goes further than what has been discussed in most academic papers in that it describes some of the implications we can draw from recognizing that there is likely more beta than alpha in hedge funds. We will discuss the possibility and reality of constructing passive, investable hedge fund indices thereof, and finally provide some remarks on the controversy of the future investment capacity for hedge funds.

The article is structured as follows: The first part gives a review of the structure of the currently available return factor models for hedge funds. The second part discusses the problems and pitfalls of hedge fund indices, before the third and fourth part provides some concrete asset based factor models for the various hedge fund strategy sectors. The fifth part discusses how one can construct real benchmarks and possibly passive and investable hedge fund indices. The subsequent two sections discuss the future of hedge funds alphas and the entire industry’s investment capacity, before we provide some concluding remarks.

Factor models for hedge fund strategies: Revisiting Sharpe’s approach

In 1992 W. Sharpe introduced a unifying framework for such style models in an effort to describe active management strategies in equity mutual funds.⁶ In his model, he

⁴ While the CAPM is considered “dead” by most academics, there are extension of it in various forms that continue to be subject of research. Further the CAPM is still in extensive use by practioners.

⁵ W. Fung, D. Hsieh, “Extracting Portable Alpha from Equity Long/Short Hedge Funds” (2004),

⁶ See “Asset Allocation: Management Style and Performance Measurement” (1992) by William Sharpe and the articles by Eugene Fama and Kenneth French “Multifactor explanations of Asset Pricing

describes a certain active investment style as a linear combination of a set of asset class indices. In other words, an active investment strategy is a linear combination of passive, i.e. long-only, buy-and-hold, strategies. The models Sharpe introduced are successful in explaining the lion's share of the performance of mutual funds.

Fung and Hsieh were the first to extend Sharpe's model to hedge funds in 1997.⁷ They employed techniques similar to those Sharpe had applied to mutual funds five years earlier, but introduced short selling, leverage and derivatives – three important techniques employed by hedge funds - into their model. The resulting factor equation would account for all hedge fund return variation that derives from risk exposure to the risk factors of various asset classes. Adding alpha to the equation, it allows us to decompose hedge fund return as:

$$\text{Hedge fund excess return} = \text{Manager's alpha} + \sum (\beta_i * \text{Factor}_i) + \text{random fluctuations}$$

Fung and Hsieh performed multifactor regressions of hedge fund returns on eight asset class indices: US equities, non-US equities, emerging market equities, US government bonds, non-US government bonds, one-month Eurodollar deposit rate, gold, and the trade-weighted value of the US dollar. They identified five risk factors (referred to as style factors), which they defined as modelling Global Macro, Systematic Trend-Following, Systematic Opportunistic, Value, Distressed Securities. They further argued that hedge fund strategies are highly dynamic and create option-like, non-linear, contingent return profiles. These non-linear profiles, they argued, cannot be modelled in simple asset class factor models. In their later research they explicitly incorporate assets with contingent payout profiles, e.g. options.⁸ Most of the studies which have followed show results consistent with Fung and Hsieh⁹. The recent literature offers an increasing number of studies around the question of common style factor exposure and contingency in payoff profile for hedge funds¹⁰.

As the formula above describes, we infer the hedge funds' alphas by measuring and subtracting out the betas times the beta factors. We can look at alpha as the "dark matter" of the hedge fund universe. It can only be measured by separating everything else out and seeing what is left. In other words, alpha is never directly observable, but is measured jointly with beta. It can only be indirectly quantified by separating the beta components out. The obtained value of alpha therefore depends on the chosen risk factors. If we leave out a relevant factor in the model, the alpha will come out as fictively high. . But one can equally say that alpha is the garbage bag of the regression: We account for everything we can, and whatever is left gets put into alpha. As a consequence, some of the returns not accounted for by these models are unaccounted beta rather than alpha. Surely, an incomplete model of systematic risk factors doesn't mean those additional risk factors do not exist; only that we do

Anomalies" (1993) and "Common risk factors in the return of stocks and bonds" (1993). More information can also be found at the websites of William Sharpe, www.wsharpe.com, and Ken French, <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/>.

⁷ Fung, W., Hsieh, D., (1997).

⁸ The idea of option factors for the purpose of hedge fund modeling was already introduced in the earliest work on hedge fund models by W. Fung and D. Hsieh, (1997), and was since then discussed by many academic studies. See their recent work: W. Fung, D. Hsieh. (2002).

⁹ See e.g. the article by S. Brown and W. Goetzmann, (2003). The authors identify eight style factors, i.e. three more than Fung and Hsieh in their research.

¹⁰ See W. Fung, D. Hsieh, (2003); (2001); (2001); " (2002); V. Agarwal, N. Naik, " (2000); D. Capocci, G. Hübner, (2004).

not yet know how to model them. To draw another image from astronomy, the outer planets of our solar system existed and exerted their gravitational pull long before we had telescopes sensitive enough to see them. The formula above on hedge fund returns should actually read as follows:

$$\text{Hedge fund return} = \text{Manager's alpha} + \sum (\beta_i * \text{Factor}_{i(\text{modelled})}) + \sum (\beta_i * \text{Factor}_{i(\text{unmodelled})}) + \text{random fluctuations.}$$

A simple example illustrates the problem: Consider a put writing strategy on the S&P 500, or equivalently the covered call writing strategy represented by the Chicago Board of Trade's BXM index, i.e. we write monthly at-the-money call options on existing equity positions with one month maturities. On regressing the BXM index against the S&P 500 over a period of 11 years from 1994 to 2004 we obtain a statistically significant alpha of around 0.4% per month, or almost 5% p.a. There is surely not much true skill driven alpha in writing put options on equities¹¹. All the 0.4% is what we refer to as spurious or "phantom" alpha, which results from the imperfect specification of the chosen model (regression against the S&P 500). So we should not confuse pure manager skill with an imperfect model. This is a common problem of multi-factor models in the literature which claim to proof high alphas. We must therefore always take any statistics of alpha with a grain of salt.

The problem with hedge fund indices

There is some more bad news for alpha: Hedge fund databases and thus the indices constructed thereof are subject to various biases which make their returns and thus the obtained alpha in a regression analysis based on these indices look bigger than they really are.¹² The lack of transparency and uniform reporting standards in the hedge fund industry are disreputable sources of measurement errors that plague any hedge fund performance analysis. The most important of these are the survivorship and the backfilling bias. The consensus view of studies on this subject is that these effects account for at least 3-4% of the reported hedge fund out-performance. A recent study by B. Malkiel and A. Saha gives an idea about the performance upward biases in hedge fund indices¹³.

There is in fact little widely published data on historical hedge fund performance, so industry analysis relies mostly on aggregated returns as provided by a dozen of different index providers which differentiate hedge fund performance across the various strategy sectors. Although these indices constitute an important tool for comparison and possibly benchmarking within and outside the hedge fund industry, measuring manager performance, classifying investment styles, and generally creating a higher degree of transparency in this still rather opaque hedge industry, the results of these efforts vary significantly between providers and depend more on "committee decisions" regarding index construction criteria - such as asset weighting, fund selection and chosen statistical adjustments - than on objectively determined rules. Although this is also somewhat of a problem in traditional asset class indices, it is severely exacerbated in the hedge fund space by the diverse, dynamic and opaque nature of the hedge fund universe.

¹¹ Writing put options and investing the collateral in cash is identical to writing covered calls, a property that is known as "put call parity in option theory.

¹² See the discussion in chapter 7 and chapter 9 in L. Jaeger "Through the Alpha Smoke Screens: A Guide to Hedge Fund Return Sources" (2005).

¹³ B. Malkiel, A. Saha, "Hedge Funds: Risk and Return", Working Paper (2004).

The built-in flaws of existing indices have as much to do with the built-in complexities of hedge funds as with any fault of the index developers. It is simply more difficult to create unambiguous index construction guidelines for the heterogeneous hedge fund universe. In particular, while the construction of traditional asset class indices rests on the reasonably well founded assumptions that the underlying assets are homogenous, and that the investor follows a “buy and hold” strategy, hedge funds are diverse and subject to dynamic change. In traditional asset classes, the average return of the underlying securities in an index has a strong theoretical basis. It is constructed to be the return of the “market portfolio,” which is the asset-weighted combination of all investable assets in that class or a representative proxy thereof. According to asset pricing theory – e.g. Sharpe’s Capital Asset Pricing Models (CAPM) - this market portfolio represents exactly the combination of assets with the optimal risk-return trade-off in market equilibrium. It is therefore not surprising that traditional equity indices became vehicles for passive investment only after the development of a clear theoretical foundation in the form of the CAPM¹⁴. Traditional indices are designed to capture directly a clearly defined risk premium available to investors willing to expose themselves to the systematic risk of the asset class. So an investor in the S&P 500 index knows exactly what he is getting; broad exposure to the risks and risk premia of the US large cap equities market. In other words, there exists a general equilibrium model. However, such a model is still missing for the asset class hedge funds.

The standard way to construct a hedge fund index has so far been to use the average performance of a set of managers¹⁵. However, indices constructed from averaging single hedge funds inherit the errors and problems of the underlying databases. Therefore they face several performance biases that limit the usefulness of the result¹⁶. These biases include (but are not limited to):

Survivorship: The survivorship bias is a result of unsuccessful managers leaving the industry, thus removing unsuccessful funds ex post from the representative index. Only their successful counterparts remain; creating a positive bias. In the most extreme case this is like lining up a number of monkeys, let them trade in the markets, take out all those that lost money, and then checking the performance of the

¹⁴ It is worth noting here that equity indices remained almost solely performance analysis tools rather than investment vehicles for many years. The first asset weighted index tracker fund (on the S&P index) started in 1973, only about five years after the CAPM became broadly accepted. The very first tracker fund was launched in 1971 and was equally weighted (on the NYSE). The problem with equally weighted indices is that they require constant rebalancing to maintain those weightings, and in the pre-1975 period (i.e. prior to deregulation of stock commissions) such rebalancing was extremely costly. Wells Fargo launched a cap-weighted tracker fund in 1973 which enabled them to reduce transactions costs. Some argue that the predominance of the S&P500 as a benchmark owes more to the ease of replication than an inherent confidence in the theoretical justification for cap-weighting, see Schoenfeld’s book “Active Index Investing” (2004)

¹⁵ Indices based on average performance of a set of managers have generally well known pitfalls, already in traditional asset classes. See the article by Jeffrey Bailey “Are Manager Universes Acceptable Performance Benchmarks,” Spring 1992.

¹⁶ Most of these issues are well known by practitioners and are discussed in details in chapter 9 of L. Jaeger “Through the Alpha Smoke Screens”. A good overview of the problems can be found in A. Kohler, “Hedge Fund Indexing: A square Peg in a round hole”, State Street Global Advisors (2003). See also “Hedge Fund Indices” by G. Crowder and L. Hennessee, Journal of Alternative Investments, (2001); “A Review of Alternative Hedge fund Indices.” by Schneeweis Partners (2001); “Welcome to the Dark Side: Hedge Fund Attrition and Survivorship Bias over the Period 1994-2001” by G. Amin et al. (2001).

rest. The survivors may all be in good shape, but they hardly represent the performance of the entire original group! Many hedge fund databases only provide information on currently operating funds, i.e. funds that have ceased operation are considered uninteresting for the investor and are purged from the database. This leads to an upwards bias in the index performance, since the performance of the disappearing funds is most likely worse than the performance of the surviving funds¹⁷. Consensus estimates about the size of the survivorship bias in hedge fund databases vary from 2% to 4%. We note that hedge fund indices are only subject to this bias to the extent that they are constructed after the fact/inception of the index. Today index providers do not restate index returns on a going forward basis as managers drop in and out of their database. Index users should only use 'live' index data rather than all historical pro forma data.

Backfilling: A variation of the survivorship bias can occur when a new fund is included into the index and his past performance is added or "backfilled" into the database. This induces another upward bias: New managers enter the database only after a period of good performance, when entry seems most attractive. Since fewer managers enter during periods of bad performance, bad performance is rarely backfilled into the averages¹⁸. Again, hedge fund indices are only subject to this bias to the extent that they are constructed after the fact/inception of the index.

Selection: Unlike public information used to compose equity and bond indices, hedge fund index providers often rely on hedge fund managers to voluntarily and correctly submit return data on their funds. Hedge fund managers are private investment vehicles and are thus not required to make public disclosure of their activities. Some bluntly refuse to submit data to any index providers. This "self-selection bias" causes significant distortions in the construction of the index and often skews the index towards a certain set of managers and strategies on a going forward basis. Sampling differences produce much of the performance deviation between the different fund indices. Hedge fund indices draw their data from different provider, the largest of which are the TASS, Hedge Fund Research (HFR) and CISDM (formerly MAR) database. These databases have surprisingly few funds in common, as most hedge funds report their data – if at all – only to a subset of the databases. Counting studies have shown that less than one out of three hedge funds in any one database contributes to the reported returns of all major hedge fund indices¹⁹.

Autocorrelation: Time lags in the valuation of securities (especially for less liquid strategies like Distressed Securities) held by hedge funds may induce a smoothening of monthly returns which leads to volatility being significantly underestimated. Statistically this effect expresses itself by significant autocorrelation in hedge fund returns (as will be shown below).

Ironically, the theoretical and practical problems described above do not disappear when the index is designed to be investable. Some problems are actually

¹⁷ The survivorship bias is also well known in the world of mutual funds, see for example the paper by S. Brown et al., "Survivorship Bias in Performance Studies" (1992).

¹⁸ R. Ibbotson estimates this bias to account for a total of up to 4% of reported hedge fund performance (Presentation at GAIM conference 2004). See also: Brown, S, Goetzmann, W., Ibbotson, R., "Offshore hedge funds: Survival and performance 1989-1995" (1999). A recent estimate of the backfilling bias is given by B. Malkiel et. al in their paper "Hedge Funds: Risk and Return" (2004) where the backfilling bias is estimated in the same region as by Ibbotson.

¹⁹ See the study by W. Fung and D. Hsieh, "Hedge Fund Benchmarks: A Risk Based Approach" (2004)

exacerbated. A prerequisite for creating an investment vehicle is that the underlying managers provide sufficient capacity for new investments. This creates a severe selection bias, as hedge funds at full capacity (closed) are a priori not considered in the index. In traditional assets, an investor in the Dow Jones Industrial Average Index does not need to worry that IBM is closed for further investment²⁰. But for hedge fund indices, capacity with top managers is a main issue. There is a clear trade-off between making an index representative and making it investable. Fig. 1 shows the divergence of various Hedge Fund Research investable indices versus their non-investable counterparts since inception of the former. The deviation is eye-catching: Let us just have a specific look at the Equity Hedge indices. The average monthly underperformance of the HFRX, the investable counterpart of the HFRI index, to the HFRI index is 62 bps, which translates into an average annual underperformance of 7.7%! We conjecture that this is about selection bias more than survivorship bias.

Investable indices depend directly on the services of particular “access providers”. The selection of the index participants is biased towards the access these service providers have to various hedge funds. This “access bias” can lead to a severe distortion in the index. The investment capacity of hedge fund managers (at least those which are actually in a position to provide persistent alpha) is a scarce resource, for which investable index providers must compete with other investors, e.g. funds of funds. An investor in a traditional S&P 500 index fund does not have to worry that stocks in IBM will not be available for purchase. But for an investable hedge fund index, availability of specific funds is indeed an issue (as for any other investor). In such non-public markets as those in which hedge funds do their offering, access is not determined by market price, but by the investors’ ability to get and keep direct access to the individual fund manager. Often this is determined by personal relationships and other “soft factors”. Therefore the distinction between indices and regular fund of funds disappears upon a closer look for most index providers²¹. The indices struggle for capacity, must perform due diligence on hedge fund managers, and have similar subjective means to select and assign weights to hedge funds. It is thus not surprising that they often charge similar levels of fees as funds of funds and in almost all cases actually also operate as such. We can essentially identify them as disguise fund of funds that have discovered the marketing value of the “index” label²².

²⁰ To be more precise, IBM stocks are in fact “closed for further investments” as there are only a finite number of shares available (assuming no capital increase). In this way they actually resemble closed hedge funds. However, any investor who desires can freely purchase IBM shares in the secondary markets (stock markets) due to its high degree of liquidity (that is what stock markets are all about). In this sense the comparison serves us well here.

²¹ The distinction between investible index providers and fund of funds is/should be about systematic methodology and goals for manager selection. Most index providers have virtually no selection methodology, and to that extent they are just fund of funds. Those that do have well founded methodologies that are implemented can, without demurring, be called indices. The biggest problem really is that the index provider and the asset manager are in fact identical—this is unlike the case for US Equity Indices, but not unlike the case for the most well regarded bond indices (e.g. Lehman).

²² One important difference between the index provider and a fund of hedge funds remains, though: The fund of funds manager is actively searching for alpha and trading talent, which justifies the comparably high fee level charged. He is not in the business of “averaging the alpha,” an undertaking which almost by construction will lead to lower results in the case of hedge funds. Note that alpha extraction is on a global scale a “zero sum game”.

They currently offer neither low fee structures nor the clearly defined risk profiles comparable to a passive index fund in traditional asset classes²³.

The true test of whether a hedge fund index is a valid investment vehicle is whether there is a secondary market for hedge funds, whether one can construct derivatives from it and whether it can be sold short. The possibility of short selling and constructing synthetic positions based on derivatives (in a cost efficient way) creates the prospect of arbitrage opportunities using the hedge fund indices. Ironically such arbitrage opportunities would most likely be exercised by hedge funds, in a sort of Klein bottle of investments that contain themselves. Whether or not such trades emerge will eventually prove whether hedge fund indices can sustain market forces, which ultimately enforce an arbitrage-free market equilibrium. Today, there is an active market for structured products referencing hedge fund indices, including delta one products that allow investors to synthetically short some of the investable hedge fund indices.

Modelling hedge fund returns – a first simple example

Fig. 2 provides a first insight into how a combination of simple systematic strategies each of which track particular “beta factors” (risk premia) tracks the performance of a multi-strategy hedge fund portfolio. It displays the return of an equally weighted combination of three simple strategies, each tracking different risk premia:

1. A simple trend following model on 25 liquid futures markets summarized on what is known as the “sgfi index” (Bloomberg ticker “SGFII <Index>”)²⁴;
2. The BXM index - an index defined by the Chicago Board of Trade for a simple “buy write” strategy on the S&P 500²⁵ (Bloomberg ticker “BXM <Index>”);
3. The Credit Suisse High Yield Bond Index (Bloomberg ticker “CSHY <Index>”).

There are no restrictions and limited fees for investing into these three strategies, and prices are readily available on information systems like Bloomberg. Figure 2 also displays the returns of the HFR Composite Hedge Fund Index, a broad aggregate across all hedge fund strategies, the Hedge Fund Research Fund of Funds Index, which mirrors the performance of fund of funds managers, and finally the S&P 500 index.

The return of this simple strategy combination over the 11-year period from 1996 to 2005 stands at 10.1% with a volatility of 5.6% and a Sharpe ratio of around 1. Compare this to a 11.1% return for the HFR Composite Index (volatility 7.1%, Sharpe ratio: 0.9) and 7.2% (volatility 5.9%, Sharpe ratio: 0.5) for the HFR Fund of Funds Index. Surprisingly, the performance of our simple strategy combination outperforms both hedge fund indices on a risk-adjusted basis. It even fares better than the HFR Fund of Funds index on a total return basis and has only marginally lower absolute returns than the HFR Composite Index. The fact that a combination of such simple strategies already beats hedge fund averages illustrates the key role of risk premia in

²³ The reader is referred to the following article for another discussion on the problems and pitfalls of hedge fund indices: L. Jaeger, “Hedge Fund Indices – A new way to invest in absolute returns strategies?”, (June 2004).

²⁴ See L. Jaeger et al., “Case study: The sGFI Futures Index”, Journal of Alternative Investments, (Summer 2002).

²⁵ “Buy write” refers to holding long the underlying – in this case the S&P 500 index, and simultaneously selling a call. This combination is economically identical to selling a put on the S&P500 plus holding an equivalent amount of cash.

hedge fund returns overall. This clearly justifies a deeper search into the risk premia of individual hedge fund strategies.

Regression of hedge fund returns on systematic risk factors

In the following we perform modelling of hedge fund strategies based on various regressions on systematic risk factors. For the lack of better data we must hereby rely on the publicly available hedge fund indices despite their shortcomings mentioned above. One might suggest that a better choice would be to perform the analysis on the investable (non-investable) indices as these do not come with these biases. However, as discussed above, these often lack the necessary degree of representativeness due to their own selection biases. Furthermore, their history is too short to perform a meaningful regression. And we claim that non-investable hedge fund indices themselves serve better as the dependent variables in a risk factor analysis as it seems at first sight, because their discussed short comings refer mostly to the absolute level of performance and not to their risk characteristics. While non-investable indices fail when used as absolute performance measures, they may very well do their service when it comes to describing the typical risk exposure characteristics of the diverse strategies²⁶. In other words, the biases such as survivorship and backfilling bias have their effects mostly on the y-intercept, i.e. the alpha, and less so on the sensitivities, i.e. the betas, of the regression. In order to illustrate this statement, we performed an analysis identical to the one above on extended sets of individual managers as provided by the TASS database. The thus obtained R-squares can be expected to be much lower due to the heterogeneity of hedge managers even within the same sector, but the obtained average values for the sensitivities are generally quite similar. Fig. 3 illustrates this for the case of Long/Short Equity managers, where we display the histograms of the obtained factor sensitivities in our regression analysis for 483 Long/Short Equity managers in the period from 1998 to 2004. These results should be compared to the results in the first row in the following Table 1.

Table 1 summarizes the results of a multifactor regression on the various hedge fund strategy sector indices provided by the data provider Hedge Fund Research (HFR). Returns are calculated on monthly data as geometric averages (cumulative returns) of the log-differences of consecutive (monthly) prices. Further the risk free rate of return was explicitly subtracted from all independent as well as the dependent variables, evidently with the exception of spread factors (as a risk free rate we chose US 3 month Libor). Note that the regression models include the AR(1) factor (the autocorrelation factor, which is the one-month lagged time series of the dependent variable) as independent variable where significant. The reason for this is simply that lagged marking of asset in several hedge fund strategies prices do not adjust instantly to changing prices of the underlying instruments but with a delay, either because the underlying markets they trade in are less liquid or because they want to smooth their reported returns over time, or, as been hypothesized elsewhere, active smoothing of returns by hedge fund managers²⁷.

Overall, the set of factors captures a large percentage of the hedge fund return characteristics, which expresses itself in the high R^2 values taking a value of 60% on

²⁶ Which is actually what linear regression models do, they explain variance, not absolute return.

²⁷ A thorough discussion of the autoregressive factor can be found in Getmansky M., Lo, A. W., Makarov, I., "An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns" (2004). See also the paper by C. Asness et al, "Do hedge funds hedge" (2001).

average. But at the same time this means that although we can explain a substantial part of the variation of hedge fund returns by these factor models, a substantial part is still missing. Furthermore, the regressions are much more successful at explaining some hedge fund strategies than others. They do well at explaining Long/Short Equity, Short Selling, and Event Driven strategies. On the other hand, they do a poorer job with the strategies Equity Market Neutral, Merger Arbitrage, and Managed Futures. We realize that hedge funds earn a substantial part of their returns by taking systematic risks that our statistical methods allow us to measure. But the nature of these risks often diverges from the standard notion of systematic (broad market) risk. In the case of equity risk factors, it is often small cap risk (Russell 2000), non-linear risk (convertible bonds, BXM), or default risk (high yield, emerging markets) rather than the risk of the overall stock market. In the case of bond market risks, it is specifically credit risk that is assumed by many hedge funds (Event Driven, Distressed Debt, Fixed Income Arbitrage, Convertible Arbitrage).

Note the significance of the autoregressive term AR(1) in the regression in five out of ten strategies. We can interpret the autocorrelation shown in the results as a sign of persistent price lags in the valuation of hedge funds. This implies that simple measures of risk like Sharpe ratio, volatility, correlation with market indices etc. significantly underestimate the true market risk in hedge fund strategies. Indeed positive autocorrelation has two effects: it drives down estimated volatility and it means that suddenly changing market conditions and shocks – as measured by the risk factors – distribute over several periods. The AR(1) factors thus measures some lagged beta. Excluding this factor would cause some unaccounted beta to be misinterpreted as alpha.

The regression results discussed here above merit a more detailed look at some of the statistics we obtained, specifically on the stability of our models, a subject which is surprisingly little covered in the literature. For this purpose we performed a CUSUM test which is designed to test whether the obtained regression models are stable to any statistically significant degree. The CUSUM test considers the cumulated sum of the (normalized) recursive residuals w_t .

$$W_t = \sum_{r=K+1}^{r=t} \frac{w_r}{\hat{\sigma}},$$

(where the denominator displays the predicted standard deviation of the error term of the regression).

In order to perform the test W_t is plotted as a function of the time variable t . The null hypothesis of model stability can be rejected when W_t breaks the straight lines passing through the point $(K, +/ - a(T-K)^{1/2})$ and $(K, +/ - 3a(T-K)^{1/2})$ where a is a parameter dependent on the chosen level of significance. Fig. 4 displays the cumulated residuals for all models. We observe that for none of our models do the cumulated residuals W_t break the confidence levels. Therefore the null hypothesis of model stability cannot be rejected for any of our models.

A second test for model stability is to plot the obtained factor sensitivities over time in a rolling regression. We equally performed this analysis, and results equally indicate a generally high degree of stability of these factors. Fig. 5 shows the results for all our strategies.

Mimicking hedge fund strategies – Can we create better indices?

The obvious question arises: Can we use the insights given by the models and the factor exposure discussed above to create better benchmarks? These would aim at mimicking the particular hedge fund strategies, and possibly constitute investable alternatives to the currently offered hedge fund indices (a provocative thought which we already hinted at in Fig. 2). The very goal would be accurately separate systematic risk exposure from true manager alpha. The former constitutes what an index is all about while the latter by definition should not be part of an index/benchmark.

The idea of using strategy replications to model hedge fund returns in a factor model setting was developed in a paper by Fung and Hsieh in 2001 for Managed Futures strategies.²⁸ Fung and Hsieh modelled the performance of a generic trend-following strategy using look-back straddles. Since then they and others have applied this type of modelling to a variety of other hedge fund styles,²⁹ including Merger Arbitrage,³⁰ Fixed Income Arbitrage,³¹ and Long Short Equity.³² The hedge fund firm Bridgewater, for example, has conducted some simple but interesting research along these lines.³³ In most of these studies the authors used simple trading strategies for modelling Managed Futures, Long/Short Equity, Merger Arbitrage, Fixed Income Arbitrage, Distressed Securities, Emerging Markets, and Short Selling strategies and generally reached good correspondence with the broadly used hedge fund sub-indices of the corresponding strategy sector.

In the following we calculate the performance of a strategy which invests directly into the factor exposures taken from the regression, i.e. we explicitly calculate the cumulative returns

$$\text{Return}(t) = \sum (\beta_i * \text{Factor}_i(t)).$$

The factors chosen for this analysis are the same as in the regression above. We refer to these returns as the “Replicating Factor Strategy” returns (in the following referred to as simply “RFS” returns) and compare them to the realized returns displayed by the corresponding hedge fund indices. In order to avoid the problem of data mining and in-sample over-fitting, the factors chosen for the RFS were calculated on a rolling looking forward basis. To be precise, the RFS returns in a given month were calculated using factors obtained by a regression over data for the

²⁸ See W. Fung, D. Hsieh, “The Risk in Hedge Fund Strategies: Theory and Evidence from Trend-Followers” (2001).

²⁹ See W. Fung, D. Hsieh, “The Risk in Hedge Fund Strategies: Alternative Alphas and Alternative Betas” in L. Jaeger (ed.), “The new generation of risk management for hedge funds and private equity investment” (2003).

³⁰ M. Mitchel, T., Pulvino, “Characteristics of Risk in Risk Arbitrage” (2001).

³¹ W. Fung, D. Hsieh, “The Risk in Fixed Income Hedge Fund Styles” (2002).

³² W. Fung, D. Hsieh, “The Risk in Long/Short Equity Hedge Funds” (2004); V. Agarwal, N. Naik, “Performance Evaluation of Hedge Funds with Option-Based and Buy-and-Hold Strategies” (2003).

³³ See the publication by G. Jensen and J. Rotenberg “Hedge Funds Selling Beta as Alpha” (2003).

previous five years ending with the previous month. The RFS are in spirit similar to what Jensen et al.³⁴ describe as a generic replication of hedge fund strategy with the difference however that the chosen factors/substrategies are explicitly modelled in the regression set up.

The results for the most recent two years (since inception of the investable indices) are rather astonishing: The cumulative replicating strategy's returns are often superior to the returns of the hedge fund indices, especially when considering their investable versions. For the latter performance of the RFS is better for every single strategy sector with the exception of the Distressed strategy.

Interpreting our results leads us to a schematic illustration of where hedge fund returns come from. A long-only manager (represented by the left bar) has two sources of returns: the market exposure and the manager excess return, his "alpha" (which is negative for most managers in this domain). The difference between long-only investing and hedge funds is largely that the hedge fund will hedge away all or part of the broad market exposure. In order to achieve this risk reduction, the hedge fund manager employs a variety of techniques and instruments not typically used by the long-only fund manager including short selling and the use of derivatives. This results in what appears as a "pure alpha" product with low expected returns and low expected risk. But in order to be attractive as a stand-alone investment, the hedge fund manager has to conform to the market standard for return. This leads him to scale the risk by using leverage, which provides the desired magnification of return and risk. In this magnified configuration, systematic elements of risk and return that before were hidden in the "Alpha" are suddenly large enough to be analysed separately. In other words, we now have the necessary magnifying glass to separate out the "beta in alpha's clothing." We estimate that up to 80% of the returns from hedge funds originate as the result of beta exposure (i.e. exposure to systematic risk factors) with the balance accounting for manager skill based alpha (or not yet identified risk factors).

In the following we discuss our results for the individual strategy sectors, the summary of which is presented in Table 2 in comparison with the investable and non investable indices from Hedge Fund Research.

Long/Short Equity

Most Long/Short Equity managers have exposure to both the broad equity market and particularly to small cap stocks. Managers may find it easier to find opportunities in a rising market, and it may also be easier to short sell large cap and buy small cap stocks. Our risk factor model in Table 1 confirms these results. The most significant factors are related to broad equity and small cap equity markets. Fung and Hsieh obtain similar results in a specific study on the Long/Short Equity strategy³⁵. They choose as independent variables the S&P 500 index and the difference between the Wilshire 1750 index and the Wilshire 750 index as a proxy for the small cap risk factor. We obtained very similar results (having chosen the Russell 2000 and Russell 1000 for the calculation of the small cap spread).

³⁴ G. Jensen and J. Rotenberg "Hedge Funds Selling Beta as Alpha" (2003), updated in 2004 and 2005.

³⁵ See W. Fung, D. Hsieh, "The Risk in Long/Short Equity Hedge Funds" (2004).

However, a closer look reveals that the exposure of Long/Short Equity hedge funds has a strongly non-linear profile. This non-linear exposure is reflected in the fact that the most explanatory independent variable is a convertible bond index³⁶. Apparently, this profile models the Long/Short Equity strategy well: Less participation on the upside, protection on the downside to a certain point, but with more expressed losses in a severe downturn of the equity markets (when convertible bonds lose their bond floor). The substitution of an equity factor with a convertible bond factor thus yields a better model than a simple equity factor³⁷. However there is another equity related factor that comes into play: Hedge funds tend to decrease their exposure in falling equity markets and increase it in rising markets, similar to a “Constant Proportion Portfolio Insurance” strategy often employed in capital protected structures. We simulate this behaviour by including such a CPPI factor based on the rolling 12 month performance of the S&P 500. Fig. 7 presents the performance of the RFS next to the HFR non-investable (HFR1) and the investable versions of HFR (HFRX) and S&P indices since inception of the HFRX (inception of the S&P index occurred later, and at its inception it was taken to the same level as the HFRX in the graph). The chart confirms what the numbers already indicated: We can very well replicate the performance of the average Long/Short Equity manager in the index with a RFS model with similar performance and volatility. The RFS performs along the HFR1 index despite some alpha displayed in Table 1. Fig. 17 sheds some light on this discrepancy: Table 1 displays the average alpha over the regression period which as Fig. 17 indicates declines quite rapidly over time. Fig. 7 in contrast only matches the most recent performance since 2003. There is only little alpha shown by Long/Short Equity managers in the most recent period as Fig. 17 indicates. Finally, the RFS outperforms both investable indices (HFRX and S&P) significantly.

Equity Market Neutral

Equity Market Neutral strategies aim at zero exposure to specific equity market factors. Correspondingly, the model in Table 1 shows only a small (however statistically significant) exposure to broad equity markets. However, the results indicate that the Equity Market Neutral style carries sensitivity to the Fama-French momentum factor UMD and the value factor (the spread of the MSCI value and growth indices). The R^2 value of the regression for Equity Market Neutral comes out lowest for all strategy sectors next to the Managed Futures. In other words, simple linear models fall short of explaining a significant part of the variation of returns for this hedge fund style. However, to mix the right combination of systematic risk exposures of Equity Market Neutral strategies right, we must distinguish two distinctly different sub-styles of this strategy. The one (often system based) approach buys undervalued stocks and sells short overvalued stocks according to a value and momentum based analysis. The second more short term oriented approach (also referred to as “Statistical Arbitrage”) trades in pairs based on a statistical analysis of relative performance deviation of similar stocks. Both styles naturally have a different exposure to the factors examined here.

³⁶ The convertible bond index primarily serves as a proxy for high tech and small cap stocks. If we include the S&P 500 and Russell 2000 index a lot (but not all) of the explanatory power of the convertible bond index goes away.

³⁷ We would like to note here, however, that the substitution of the Convertible factor with a straight equity risk factor such as the S&P 500, yields R-squares which are only about 10% below the values reported here. The convertible bond index thus can be considered as a proxy for small cap (an possibly Telecom/Technology) exposure.

Fig. 8 confirms what the numbers in Table 1 indicate: The RFS underperforms the HFRI index by some margin reflecting the positive alpha in Table 1. However, it outperforms the HFRX investable index significantly.

Short Selling

The main exposure of the Short Selling strategy is, quite obviously, being short the equity market. Interestingly, the exposure to the broad equity markets can best be modeled with the same factor as for the Long/Short equity managers, the Convertible Bond Index. This indicates the same type of non-linear exposure as for the Long/Short Equity strategy, however with the signs inversed. The strategy displays positive sensitivity to value stocks with as measured by the spread between the MSCI value and growth indices. The alpha value for Short Selling strategies stands at around 4-5% p.a. This indicates that the short side does offer some profit opportunities, possibly explained in part by most investors being restricted from selling short. However, the alpha of this strategy must be high in order for the strategy to generate any profits at all. This is because from the perspective of risk factor exposure, shorting the equity markets starts off with an expected negative 4-7% return (long term performance of the equity markets minus short rebate for the short positions). As a result Short Selling is the only hedge fund strategy with negative past performance over the last 15 years. This is also reflected in Fig. 9 for the more recent period. We observe that the Short Selling strategy can be well replicated by the RFS model.

Event Driven

Event Driven hedge funds constitute an ensemble of various investment strategies around company specific events including restructuring, distress and mergers. According to our factor model in Table 1 the average Event Driven strategy comes with a rather simple exposure to the broad equity market, small cap stocks and the high yield bond market. Further the AR(1) factor indicates autocorrelation in returns reflecting liquidity risk and possible lagged pricing of the underlying securities. Our model explains an astonishing 80% of the variation of Event Driven returns. Alpha is the highest for any strategy in the hedge fund universe with roughly 5% p.a. over the analyzed period. This is also reflected in Fig. 10, where we see that the RFS model yields roughly about two thirds of the return of the Event Driven managers in the HFRI index. However, again, the RFS outperforms the HFRX and S&P investable index version significantly.

Distressed Securities

Distressed Securities strategies come with a simple set of exposures to credit, equity, particularly small cap equity, and liquidity risks. These are exactly the factors which show up in Table 1. The AR(1) factor bears the largest sensitivity, reflecting the low degree of liquidity offered in Distressed Securities investing. A lack of regular pricing and valuation induces autocorrelation in the return streams. The partly rather illiquid strategies closely resemble the return sources of private equity investment. The investor provides an important funding source for companies without access to traditional capital sources during important phases of their development; usually times of distress. In contrast to investors in regular stocks, an investor in distressed debt or equity just like a private equity investor has no direct access to his capital for several years. He is further exposed to uncertainty about the size and timing of future cash flows.

Not surprisingly the level of alpha for Distressed hedge funds managers is around 3-4% p.a. which is along with its peers in other Event Driven sectors (e.g. Merger Arbitrage) among the highest in the hedge fund industry. This is also reflected in Fig. 11, where we see that the RFS model yields roughly about half of the return of the Distressed managers in the HFRI index. Even the investable HFRX index outperforms the RFS.

Merger Arbitrage

In their seminal paper on the Merger Arbitrage strategy, Mitchel and Pulvino³⁸ examine the conditional correlation properties of this strategy: Merger Arbitrage strategies display rather high correlations to the equity markets when the latter declines and comparably low correlations when stocks trade up or sideways. This corresponds to a correlation profile similar to that of a sold put on equities. As a matter of fact, the payout profile of Merger Arbitrage strategies corresponds directly to a sold put option on announced merger deals. This short put profile is reflected in the significance of the BXM factor in Table 1. Shorting put options provides limited upside but full participation on the downside (less the option premium). This argument extends beyond the immediate exposure to merger deals breaking up: When the stock market falls sharply, merger deals are more likely to break. In addition, a sharp stock market decline will reduce the likelihood of revised (higher) bids and/or bidding competition for merger targets. Falling stock markets also tend to reduce the overall number of mergers, which increases the competition for investment opportunities and may thereby reduce the expected risk premium. The strategy therefore has a slightly positive stock market beta, however strongly non-linear. This overall exposure profile to equity markets comes more from the correlation between the event risk and the market than from the individual positions. Mitchell and Pulvino calculated the historical track record of a simple rule-based merger arbitrage strategy that at any time invests in each announced merger deal, both cash and stock-swap, with a pre-specified entry and exit rule.³⁹ They conducted this calculation for 4,750 merger transactions from 1963 to 1998. The hedge fund manager Bridgewater performed a very similar study but constrained themselves to the ten largest mergers at any point in time. In both cases the resulting simulated returns came very close to the returns of the Merger Arbitrage hedge fund indices (HFR and Tremont). We included a strategy which focuses on investing exactly along the Mitchell/Pulvino study, the publicly available "Merger Fund"⁴⁰.

Our regression shows what we expected, exposure to the equity markets, in particular the small cap segment (furthermore the value sector), the BXM index and the Merger Fund. However, the explanatory strength of the model is not that high (considering that these factor should very well reflect what the strategy is about). Just as with other Event Driven strategies the alpha value is above average for this strategy with around 4% p.a. However, a comparison with the performance of the RFS in Fig. 12 shows that the skill based component of returns has declined in recent years, as the RFS tracks the performance of the HFRI Merger Arbitrage rather closely. Again, the RFS outperforms the investable version of the HFR index by a safe margin.

³⁸ See M. Mitchel and T. Pulvino, "Characteristics of Risk in Risk Arbitrage" (2001).

³⁹ See M. Mitchel and T. Pulvino, "Characteristics of Risk in Risk Arbitrage" (2001).

⁴⁰ Bloomberg ticker: MERFX US Equity.

General Relative Value

Relative Value strategies—represented here by Fixed Income Arbitrage and Convertible Arbitrage – have three types of systematic exposure. They first capitalize on price spreads between two or more related financial instruments which often represent a compensation for particular risks such as credit risk, interest rate term structure risk, liquidity risk, or exchange rate risk. Secondly, they provide liquidity and price transparency in complex instruments employing proprietary valuation models to value complex financial instruments. Related returns can be referred to as liquidity and “complexity” premia. The latter is related to the risk of mis-modeling the complexity of the underlying financial instrument. The hedge fund manager is short an option which turns strongly into the money when his valuation model is inaccurate. Finally, Relative Value Hedge fund managers have a preference for negatively skewed return distribution, where steady but small gains are countered with rare but large losses. In other words, the managers are short some sort of volatility, which makes the return profile resemble the payout profile of a short option position.

Fixed Income Arbitrage

Fixed Income Arbitrage strategies often expose themselves to a combination of liquidity, credit and term structure risks, e.g. through credit barbell strategies (long short-term debt of lower credit quality and short long term government bonds), yield curve spread trades, or on-the-run versus off-the-run treasury bond positions. Exposure to credit risk, convertible bonds and emerging market bonds securities are most prevalent, as Table 1 indicates. The significance of the AR(1) term indicates autocorrelation in returns signaling lagged pricing of the underlying securities and reflects liquidity risk. According to our factor model the alpha value for Fixed Income strategies is in the region of 2.5% p.a., and the model explains around 41% of the variations of returns.

Fung and Hsieh⁴¹ chose another--but similar--set of factors including options on interest spreads (they call these “ABS factors”) to model various Fixed Income Arbitrage trading styles. They obtain slightly higher R^2 values than presented in our study here.

Their and our results explain why the heaviest losses of this style occurred in “flight to quality” scenarios, when credit spreads suddenly widen, liquidity evaporates and emerging markets fall sharply. Events like the summer 1998 remind us that the strategy bears a risk profile similar to a short option, with the risk of significant losses but otherwise steady returns. It is inherently difficult to model the exposure to these extreme events, as they are so rare that their true likelihood is hard to calculate. However, the hedge fund investor should nevertheless keep this exposure in mind.

Fig. 13 shows that the RFS returns cannot quite keep up with the HFRI returns coherent with in our results in Table 1.

Convertible Arbitrage

Convertible Arbitrage hedge funds are exposed to a variety of different risk factors: Credit risk, equity market and equity volatility risk, and liquidity risk. These factors – the high yield factor, convertible and equity factor, and the AR(1) factor – also appear as the relevant factors in Table 1. As for Fixed Income Arbitrage, the Convertible Arbitrage model shows a significant AR(1) terms which indicates autocorrelation in

⁴¹ See W. Fung, D. Hsieh, “The Risk in Fixed Income Hedge Fund Styles” (2002).

returns also for this strategy. This signals a lack of consistent and timely pricing of the underlying convertible securities and reflects exposure to liquidity risk and valuation risk.

To mix the right combination of these risks however, we must distinguish two distinctly different sub-styles of Convertible Arbitrage strategies. The option-based Convertible Arbitrage style simply buys the convertible bond, sells short the underlying equity and re-establishes a delta hedge frequently, a trading technique referred to as gamma-trading. This style tries to hedge out credit risk as much as possible and thus cares little about the credit markets. The second - credit-oriented - style makes an explicit assessment of the issuer's creditworthiness and takes overpriced credit risk. Both styles naturally have a different exposure to the credit markets.

Naturally, the credit-oriented sub-style of Convertible Arbitrage carries a significant exposure to credit risk, while the option-based sub-style does not. As credit risk is correlated with equity markets the second style has a less well-defined sensitivity to falling equities. Increasing volatility helps the strategy, but widening credit spreads hurt it. The option-based gamma trading style, in contrast, performs better in a volatile environment in which equities are falling, which explains the overall negative correlation of Convertible Arbitrage hedge funds to the equity markets in Table 1. Declining volatility leads this strategy to under-perform during the period of decline. The dual nature of Convertible Arbitrage hedge funds led to an interesting development in 2003 which confused some investors. In an environment of simultaneously rapidly declining credit spreads and equity volatility, credit oriented Convertible Arbitrage strategies displayed stellar performance while the gamma traders displayed disappointing returns that hovered near zero.

This divergence in style is currently not reflected in the available hedge fund indices, which makes it more difficult for factor models to capture the sensitivities of the style.

To correctly evaluate these two variants of Convertible Arbitrage, we would need a separate index for each sub-style. In a recent research paper⁴², V. Agarwal et al. separate the key risk factors in Convertible Arbitrage strategies: equity (and volatility) risk, credit risk, and interest rate risk. Consequently they design three "primitive trading strategies" to explain the returns of the strategy in terms of the key risk factors and premia captured by these strategies: positive carry, credit risk premium ("credit arbitrage") and gamma trading ("volatility arbitrage"). They investigate these factors in the US and Japanese convertible market. These factors can explain up to 54% of the return variation of Convertible Arbitrage indices.

According to our factor model the alpha value for Convertible Arbitrage Income strategies is in the region of 2% p.a, and the model explains around 65% of the variations of returns. However, we observe for the more recent period that a RFS model outperforms the HFRI Convertible Arbitrage strategy slightly with significantly less volatility as shown in Fig. 14. The outperformance becomes even more striking when considering the investable HFRX index.

Global Macro

⁴² V. Agarwal, W. Fung, Y. Loon, N. Naik, "Risks in Hedge Fund Strategies: Case of Convertible Arbitrage" (2004).

Global Macro managers of all types do better in strong bond markets, as indicated by the strong sensitivity to the bond market index shown in Table 1. Other exposures are less obvious: exposure to the risk characteristic to trend following strategies (the sGFI factor) and some non-linear exposure to the broad equity market (convertible bond factor).

The R^2 value for the regression of Global Macro comes out relatively low (50%). We assume this is due to the heterogeneity of the strategy. Global Macro trading includes a wide range of different trading approaches, and a broad index does not reflect this diversity. A manager-based analysis would be more appropriate here. More than a broad asset class based index or a generic trading strategy, it is the particular markets traded by the individual manager and his particular investment techniques that define the available risk premia and inefficiencies targeted. However, note that our model gives an alpha value of around 3% p.a. for the average Global Macro manager. This is correspondingly reflected in Fig. 15, showing an underperformance of RFS of around 3-4% p.a.. But again, the non-investable version underperforms the RFS.

Managed Futures

Managed Futures hedge funds are the main speculative agents in the global futures markets, thus capturing what we referred to as the “commodity hedging demand premium”. A simple trend following trading rule (sGFII) applied to the major global futures markets captures a large part of these returns and shows up as the most dominant term in the regression in Table 1. Several different studies have independently obtained this result.⁴³ The sGFII index is designed to model the return of trend following strategies with a simple rule based momentum approach. It is a volatility weighted combination of trend following strategies on 25 liquid futures contracts on commodities, bonds, and currencies. This index shows a 48% correlation with the CISDM trend following index, and equally a 48% correlation with the CSFB/Tremont index. Based on the regression in Table 1 the average CTA in the CISDM Trendfollower index displays negative alpha. Schneeweis/Spurgin and Jensen and Rotenberg (Bridgewater) use similar trend following indicators on a much more restricted set of contracts⁴⁴. They obtain an even higher correlation coefficient to the CSFB-Tremont Managed Futures index (71% in the case of Bridgewater) or the CISDM Managed Futures Indices (79% against the CISDM Trend following index for Schneeweis/Spurgin). The lower correlation of the sGFII index is possibly due to a comparably high exposure to commodity contracts compared to Bridgewater’s and Schneeweiss/Spurgin’s model (which overweigh the complex of financial futures contracts)

An interesting model for trend-following strategies was proposed by Fung and Hsieh. They constructed their trend-following factor using look back straddle payout profiles on 26 liquid global futures contracts and the corresponding options (across equities, bonds, currencies and commodities). A look back straddle pays the difference between the highest and lowest price of the reference asset in the period of time until maturity of the option, mimicking the payout of a trend-follower with perfect foresight.

⁴³ See L. Jaeger et al., “Case study: The sGFI Futures Index” (Summer 2002); Jensen. G., Rotenberg, J., “Hedge Funds Selling Beta as Alpha” (2003); R. Spurgin, “A Benchmark on Commodity Trading Advisor Performance “ (1999).

⁴⁴ T. Schneeweis and R. Spurgin, “Multifactor Analysis of Hedge Funds, Managed Futures, and Mutual Fund Returns and Risk Characteristics” (1998); G. Jensen and J. Rotenberg “Hedge Funds Selling Beta as Alpha” (2003).

The degree of explanatory power of their model is around $R^2=48\%$, higher than all three models described above.

Note that the Managed Futures strategy is the only hedge fund sector which displays negative alpha (albeit not at a statistically significant level). We can observe the corresponding performance pattern of CTAs compared to the RFS in Fig. 16: The performance of the RFS and the average CTA in the CISDM Managed Futures Qualified Universe Futures Index are very well in line, while the investable S&P Managed Futures index underperforms both by a significant margin.

The future of alpha

There is good reason to believe that generally the average alpha extracted by hedge fund managers is destined to decline. As a matter of fact, we can already today observe that alpha has grown smaller in size over time, as Fig. 17 indicates for the most obvious strategy, Long/Short Equity, where we display the alpha of a rolling regression over a 60 months time window. Independently from our research, the attenuation of alpha has been observed elsewhere. Fung et al. report in one of their latter research on the same phenomenon.⁴⁵ One possible explanation for this phenomenon comes quickly to mind: As more money chases a limited of market inefficiencies, those inefficiencies should decrease or even going to disappear. In other words, the capacity for alpha is limited. However, there is no good reason to believe that the global “capacity for alpha” which is ultimately a function of how many inefficiencies the average global investor (and the corresponding regulatory agencies) will tolerate actually decreased over time that dramatically. While hedge funds grow strongly and possibly have to compete harder with other “alpha chasers” they remain a rather small portion of the global investment activity. Another parallel explanation for the displayed decrease in alpha is the quality of the average hedge fund manager. The number of managers has multiplied in recent years, and it reasonable to assume that today’s low entry barriers to starting a hedge fund attracts numerous managers with a lower skill level. These tend to dilute the average performance and thus the average alpha of the entire hedge fund industry. An interesting research topic which we leave for future efforts is to test for the average alpha in the top percentile of managers.

Will the “alpha” in hedge funds disappear entirely? Probably not, but it will become harder to identify and isolate it in the growing jungle of hedge funds. However, we have seen that alpha constitutes a statistically significant variable (though decreasing over time) in most of our regression models. We might be missing explanatory variables in our models, and future modeling effort will hopefully lead us to better models to answer this question.

Another approach is to model the behavior of the alpha output of our models in changing market conditions as well as over time. Alpha might depend on market related variables other than prices which are not so easily captured in our risk based models, such as trading volume, open short interest on stocks, insider activity, leverage financing policies of prime brokers, etc. A direct dependency of the hedge fund managers’ alpha creation from these variables will lead us to a better understanding of their time variability that we empirically observe in our models. This will ultimately lead us to an understanding of the very alpha creation process of

⁴⁵ W. Fung, D. Hsieh, N. Naik, T. Ramadorai, „Hedge Fund: Performance, Risk and Capital Formation“, Preprint (2005)

hedge funds, the part of hedge fund returns which remains still in the dark for most investors. However, little effort has been put into this task so far.

The main task of the investor will be to define what he wants from hedge funds. Alpha is and will continue to be ultimately the most attractive sort of return, as it comes with no systematic risk and no correlation to other asset classes. But investors should realize both the scarcity of true alpha and the power of alternative beta. It is the power of diversification into orthogonal risk factors which will ensure that hedge funds remain broadly attractive for investors. And when it come to the hedge funds' beta there is surely a great deal larger capacity available to investors than in the case of alpha. In fact, the future growth prospects of the hedge fund industry become quite compelling considering that we are far from any limit with respect to "beta capacity" in the hedge fund industry. While the search of alpha surely remains compelling, we believe it is investment in alternative betas which will be more and more the key to successful hedge fund investing in the future.

The future of hedge fund capacity

Now that we are in a position to provide a rough breakdown of hedge funds return sources we can approach a question which lies at the heart of future hedge fund growth: the issue of capacity. For this purpose we perform a set of rather simple calculations:⁴⁶ We know that the global market capitalization of all public stocks and debt is around 88'000 billion USD (about 51'300 USD in bonds, 36'700 USD in equity).⁴⁷ Generating alpha in the global capital markets is an overall zero sum game, i.e. if hedge fund managers win this game, i.e. generate positive alpha, there must be other market participants being on the losing end. We must thus assume an average tolerance level for inefficiencies, i.e. negative alpha, by equity and bond investors world wide before competitive (or regulatory) forces step in to keep this number from getting larger. We estimate this number to be in the range of 0.25% p.a. on average across all equity and bonds investors.⁴⁸ With this number we can calculate the overall alpha in the global equity and bond market to be USD 220 billion. We must further assume that hedge funds can participate from this "alpha pie" only to a certain extent next to other professional players which are likely to be "positive alpha players" and thus compete with hedge funds for alpha (proprietary trading operations, large institutions, mutual funds – before their fees, etc.). It seems realistic to assume that hedge funds can take one fourth of that pie⁴⁹ (a proportion which might grow larger over time, however, as more players from the other "alpha parties" move into the hedge fund space). This implies that there are USD 55 billion pure alpha available to hedge funds each year. Further, assuming that hedge fund investors require a least a 15% p.a. return *gross of fees* (before management, performance, trading fees, etc.), which amounts into a net return of around 8%-10% and constitutes probably the minimum investors would require from hedge funds. This implies an overall capacity of hedge funds *based on alpha only* of

$$\text{USD 55 billion} / 0.15 = \mathbf{366.6 \text{ billion USD}},$$

⁴⁶ Note that this calculation is very similar in spirit and takes some of its concepts from the work of H. Till, "The capacity implications of the search of alpha" (2004).

⁴⁷ Source: www.fibv.com/publications/Focus0605.pdf and <http://www.imf.org/external/pubs/ft/GFSR/2005/01/index.htm>.

⁴⁸ H. Till uses another number but aggregates the overall size of the market only over the holdings of HNWI, mutual funds and institutional funds. Considering our base number of 88000 billion USD the assumptions are rather similar.

⁴⁹ The reader is invited to perform the calculation with different numbers.

about one third of the actual size of assets in the hedge fund industry. Even with different, more beneficial assumptions on the overall investor tolerance for inefficiencies and on how much hedge funds can participate in the total “alpha pie”, we would not come up with a capacity significantly higher than the current size of the industry. As a result, based on inefficiencies alone, we are not just lacking a satisfying economic explanation of hedge fund return sources, we also find ourselves in a position not being able to explain the current size of the industry!

But by now we understand that a large portion of hedge fund returns is not related to pure alpha, but rather to “alternative beta”. The analysis in our research suggests that a large part of the average hedge fund return stems from alternative beta rather than alpha. We now consider our estimate for that part to be as high as 80%. Well, this raises the bar for hedge fund capacity significantly higher. Going along with our conclusion and estimating that only 20% of the industry returns is related to pure alpha, we can calculate the capacity of the industry to be

$$366.6 \text{ billion USD} / 0.2 = \mathbf{1'833 \text{ billion USD}},$$

about twice its current size. However, as large as this number seems, it is exceeded by some of the estimates given by industry protagonists as to what level the industry will grow within the following years. How can this growth be managed considering our numbers? The answer is obvious: Only by including a larger share of alternative beta in the overall return scheme of hedge funds. Assuming that the ratio of alpha versus alternative beta becomes 10%, the capacity reaches the number of 3670 billion USD (assuming that the capacity of alternative beta is not limited at these levels, a fair assumption in our view).

Summarizing, there is indeed plenty of room for the hedge fund industry to grow, albeit only at the expense of becoming more and more beta driven. This development will inevitably occur with the future growth of hedge funds. As a matter of fact, recent performance suggests that this process has already started.

Summary and Conclusion

The key to the hedge fund ‘black box’ is the understanding that hedge funds generate returns primarily through risk premia and only secondarily by exploiting inefficiencies in imperfect markets. Conceptually hedge funds are therefore nothing really new in that just as an equity mutual fund extracts the equity risk premium, a hedge fund may try to extract various other risk premia awarded for, say, credit risk, interest rate risk or liquidity risk. The important difference however is, that the underlying risk premia are more diverse than those in traditional asset classes (which led us to refer to these premia as “alternative betas”). This insight is slowly spreading among the most sophisticated circles in the hedge fund industry. The underlying systematic risks can be readily analyzed and understood by investors, while the remaining parts of returns from inefficiencies are more difficult to describe in an unambiguous way. The risk premia available to hedge fund managers are the same as those available to other investors. However, extracting those premia in markets unfamiliar to most investors requires special expertise. Like the mining engineer who can profitably extract gold from low-grade ore that would previously have been left in the ground, skilled fund managers are simply more efficient in identifying existing risk premia, and trading with minimal risk exposure and transaction costs to extract them.

One of the pitfalls of hedge funds is that alpha and beta currently do not come separate but in an uncontrolled and perhaps undesired combination. Traditional portfolio management has developed a setting, which could equally be applicable for hedge fund investors: the “core-satellite” framework. Here, alpha generation and beta extraction are well separated - and very differently compensated. We believe hedge fund investors will want to walk down the same road. Hedge fund product providers might have to find a way to isolate and extract the alpha from the beta in hedge funds. This is the idea of “portable alpha”: Isolate alpha in one asset class and transfer it into the portfolio consisting of other types of assets. If a fund manager claims to produce alpha, why not take out the beta part of his returns with an active hedging overlay approach and keep only the alpha. A recent paper by B. Fung and D. Hsieh⁵⁰ provides some interesting insights into a possible implementation of that idea and also gives some useful estimates about size and distributional properties of the “alpha returns” for Long Short Equity strategies.

Currently available indices or benchmarks which rely on manager and peer group averages do not necessarily provide a sufficiently accurate picture of the industry or strategy sector performance due to various well known biases. The situation does not become much better when the indices are designed to be investable. At the same time, the demand and necessity of hedge fund indices for the purpose of measuring manager performance, classifying investment styles, and generally creating a higher degree of transparency is high and increasing. Some index providers actually claim that funds of funds have started to invest in investable indices to gain the desired exposure. While the authors are not aware of such behavior, they can surely not exclude that some of the less sophisticated fund of funds have bought the marketing story of the index providers. But if we acknowledge that the investable indices are no valid choice, what can we do? One way suggested in this article is to create synthetic benchmarks based on the factor exposure of hedge fund strategies to the underlying risk factors. This could potentially be a much better choice for fund of funds and other investors to gain the desired broad exposure to the hedge fund styles. At the same time these replicating factor strategies (RFS) can serve fund of funds as a benchmarking tool to judge the performance, to be more precise, the alpha generation, of their managers. First results described here and elsewhere look promising for some strategy sectors. However, a great deal of work remains to be done for other strategies. We observe that a corresponding replication of hedge fund indices by “replicating factor strategies” (RFS) lives up to the returns of the (non-investable) hedge fund strategy sector indices for some strategy sectors, in particular Long/Short Equity, Merger Arbitrage, Managed Futures, and Convertible Arbitrage. These strategies make up significantly more than 50% of the assets allocated to hedge funds! But as we emphasized in this article, these non-investable indices are actually not a good measure for hedge fund return that an investor would actually obtain on average. In contrast to the non-investable hedge fund indices the RFS can be made investable without impacting their returns. When we compare the returns of the RFS with the corresponding version of the investable indices, their outperformance becomes even more striking: The RFS actually outperform the entire range of investable indices by a safe margin with the one exception of the Distressed sector. One must wonder why this is so. The flippant but accurate answer is: fees. Taking out an average of 2% management fees and a share of 20% performance

⁵⁰ W. Fung, D. Hsieh “Extracting Portable Alpha from Equity Long/Short Hedge Funds”, Journal of Investment Management (2004).

fees for the single hedge fund manager actually eats up all and often more of the skill based returns hedge fund managers offer *on average*. We emphasize that the last two words written in italics are important: “on average”. With the inflation of new often mediocre managers average alpha has been coming down. However, we acknowledge that there continue to exist highly skilled hedge fund managers which continue to generate persistent alpha even after their (hefty) fees. It remains the skill of the experienced hedge fund investor/fund of funds to find and invest in them.

At the end of this report we would like to point out a further direction of research possibly not sufficiently covered in this research. Our analysis suggests that the factor loads of hedge fund strategies are adequately modelled as stationary. However, there is good reason to believe (and recent research provides some evidence⁵¹) that there occur structural breaks in the systematic risk exposures of hedge funds. Examples of such are easy to find: The blow up of LTCM in the summer of 1998, the burst of the stock market bubble in the spring of 2000, the turn in the equity market in March 2003. Upon a closer look, a closer look at Fig. 4 reveals some evidence for such breaks, which our analysis here does not account for. We will leave this topic for future research.

Generally, the progress recently on understanding the generic sources of hedge fund returns leads us to the conclusion that investable benchmarks constructed by a joint venture of financial engineers and quant groups based on risk factor analysis and replication has the potential to offer a valid, theoretically more sound, and cheaper alternative to the currently offered hedge fund index products offered today. It is evident that once these indices become more broadly recognized the hedge fund industry will be put upside down. This will have some further important consequences on how hedge funds are categorized by investors. So far, most consider them a separate asset class. Realizing that hedge funds regarding their exposure to systematic risk factors are conceptually not that different from traditional types of investments investors may find it conceptually easier to integrate them into their overall asset allocation.

Literature

Agarwal, V., and Naik, N., “Performance Evaluation of Hedge Funds with Option-Based and Buy-and-Hold Strategies,” Working paper (2001), published under the title: “Risks and Portfolio Decisions involving Hedge Funds”, Review of Financial Studies, 17, p. 63 (2004)

Agarwal, V., Fung, W., Loon, Y., Naik, N. „Risks in Hedge Fund Strategies: Case of Convertible Arbitrage“, Working Paper, London Business School (2004)

Amin, G., Kat, H., “Welcome to the Dark Side: Hedge Fund Attrition and Survivorship Bias over the Period 1994-2001,” Journal of Alternative Investments (Summer 2003)

⁵¹ W. Fung, D. Hsieh, N. Naik, T. Ramadorai, „Hedge Fund: Performance, Risk and Capital Formation“, Preprint (2005)

Asness, C., Krail, R., Liew, J., "Do hedge funds hedge?"; *Journal of Portfolio Management*, 28, no. 1 (Fall 2001)

Asness, C., "An Alternative Future, I & II" *Journal of Portfolio Management* (October 2004)

Bailey, J., "Are Manager Universes Acceptable Performance Benchmarks," *The Journal of Portfolio Management* (Spring 1992).

Brown, S, Goetzmann, W., Ibbotson, R., Ross, S., "Survivorship Bias in Performance Studies", *Review in Financial Studies*, Vol. 5, No. 4 (1992)

Brown, S, Goetzmann, W., Ibbotson, R., "Offshore hedge funds: Survival and performance 1989-1995", *Journal of Business*, 92 (1999)

Brown, S. and Goetzmann, W. (2003), "Hedge Funds With Style," *Journal of Portfolio Management*, 29, 101-112.

Capocci, D., Hübner, G., "Analysis of hedge fund performance", *Journal of Empirical Finance*, Vol. 11 (2004)

Crowder, G., "Hedge Fund Indices", *Journal of Alternative Investments* (Summer 2001)

Fama, E., French, K., "Common risk factors in the return of stocks and bonds", *Journal of Financial Economics*, 33 (1993)

Fama, E., French, K., "Multifactor explanations of Asset Pricing Anomalies", *Journal of Finance*, 51, p. 55 (1996)

Fung, W., Hsieh, D., Naik, N., Ramadorai, T., „Hedge Fund: Performance, Risk and Capital Formation“, Preprint (2005)

Fung, W., Hsieh, D., "Empirical Characteristics of Dynamic Trading Strategies: The Case of Hedge Funds", *The Review of Financial Studies*, No. 10, 2 (1997)

Fung, W., Hsieh, D., "The Risk in Hedge Fund strategies: Theory and Evidence from Trend-Followers," *The Review of Financial Studies*, Vol. 14, 2, p. 313 (Summer 2001) (2001).

Fung, W., Hsieh, D., "Benchmarks of Hedge Fund Performance: Information Content and Measurement Biases," *Financial Analyst Journal* (2001).

Fung, W., Hsieh, D., "The Risk in Fixed Income Hedge Fund Styles", *Journal of Fixed Income*, vol. 12, 2 (2002)

Fung, W., Hsieh, D., "The Risk in Hedge Fund Strategies: Alternative Alphas and Alternative Betas" in L. Jaeger (ed.), "The new generation of risk management for hedge funds and private equity investment", *Euromoney* (2003);

Fung, W. Hsieh, D., "Hedge Fund Benchmarks: A Risk Based Approach", Working Paper (2004)

Fung, W., Hsieh, D., "The Risk in Long/Short Equity Hedge Funds", Working Paper, London Business School, Duke University (2004)

Fung, W., Hsieh, D., "Extracting Portable Alpha from Equity Long/Short Hedge Funds" (2004), Journal of Investment Management, Vol. 2, No. 4 , p. 1-19

Getmansky M., Lo, A. W., Makarov, I., "An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns", Journal of Financial Economics, 74 (3), pp.529-610 Economics (2004)

Jaeger, L , "Through the Alpha Smoke Screens: A Guide to Hedge Fund Return Sources", Euromoney Institutional Investors (2005)

Jaeger, L., "Hedge Fund Indices – A new way to invest in absolute returns strategies?", AIMA Newsletter (June 2004)

Jaeger, L. (ed.), "The new generation of risk management for hedge funds and private equity investment", Euromoney (2003);

Jaeger, L., "Managing Risk in Alternative Investment Strategies", Financial Times/Prentice Hall (May 2002)

Jaeger, Lars, "Sources of Return for Hedge funds and Managed Futures", The Capital Guide to Hedge Funds 2003, ISIPublications (Nov. 2002)

Jaeger, L., Jacquemai, M., Cittadini, P., "Case study: The sGFI Futures Index," The Journal of Alternative Investment (Summer 2002).

Jensen, J., Rotenberg, J., "Hedge Funds Selling Beta as Alpha" Bridgewater (2003, updated 2004 and 2005)

Kohler, A., "Hedge Fund Indexing: A square Peg in a round hole", State Street Global Advisors (2003).

Malkiel, B., Saha, A., "Hedge Funds: Risk and Return", Working Paper (2004)

Mitchel, M, Pulvino, T, "Characteristics of Risk in Risk Arbitrage", Journal of Finance, Vol. 56, No. 6, p. 2135 (2001)

Schneeweis, T., "A Review of Alternative Hedge Fund Indices." Schneeweis Partners (2001);

Steven A. Schoenfeld, "Active Index Investing", Wiley Publications (2004)

Spurgin, R., "A Benchmark on Commodity Trading Advisor Performance", Journal of Alternative Investments (Fall 1999)

Sharpe, W., "Asset Allocation: Management style and performance measurement", Journal of Portfolio Management, no.18, 2 (Winter 1991)

Till, H., "The capacity implications of the search of alpha", AIMA Newsletter (2004)

Figures:

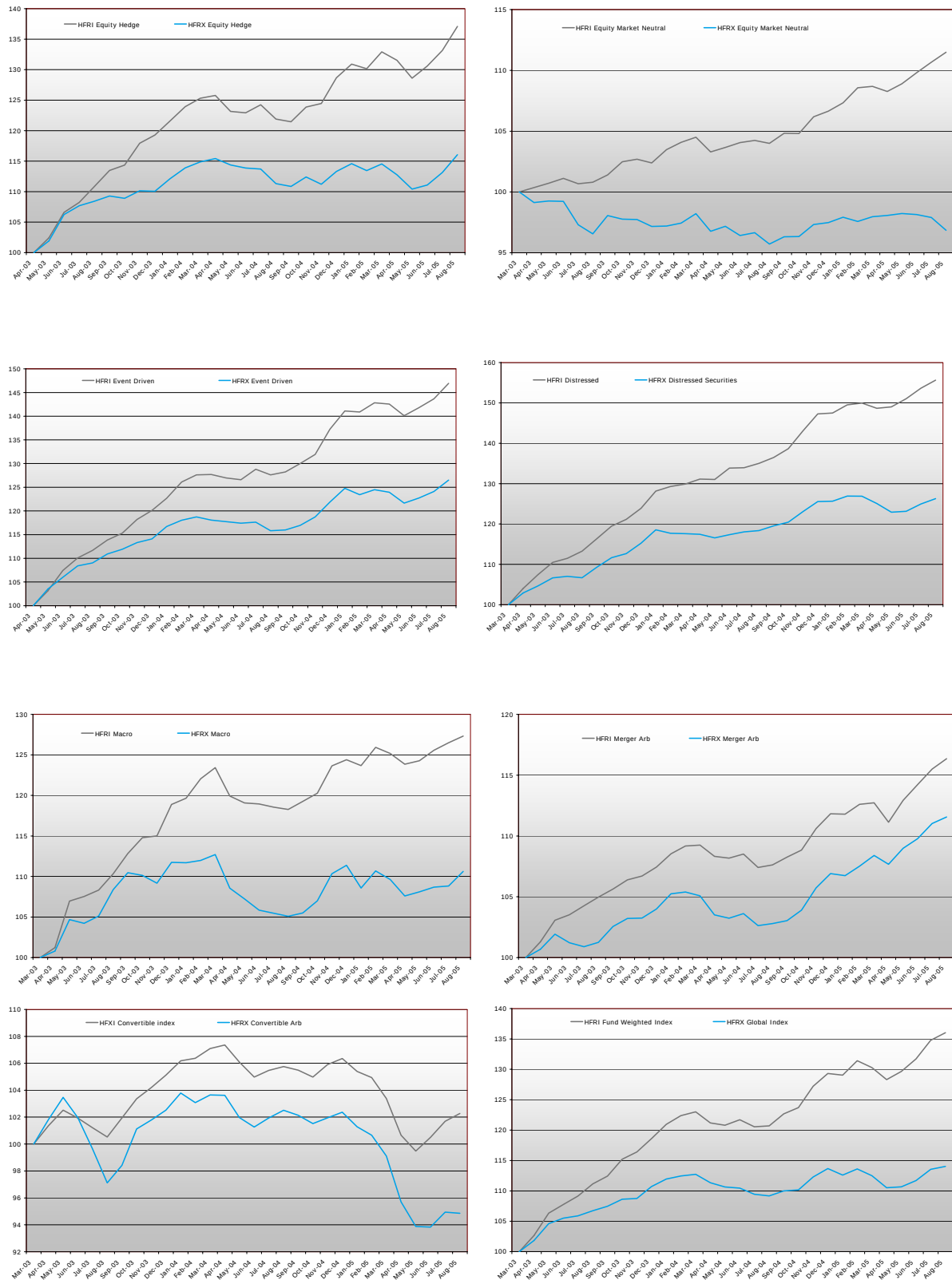


Fig. 1: Comparison of cumulative performance for the HFR investable indices versus their non-investable counterparts since inception of the former. The last graph shows the index referring to global hedge fund industry.

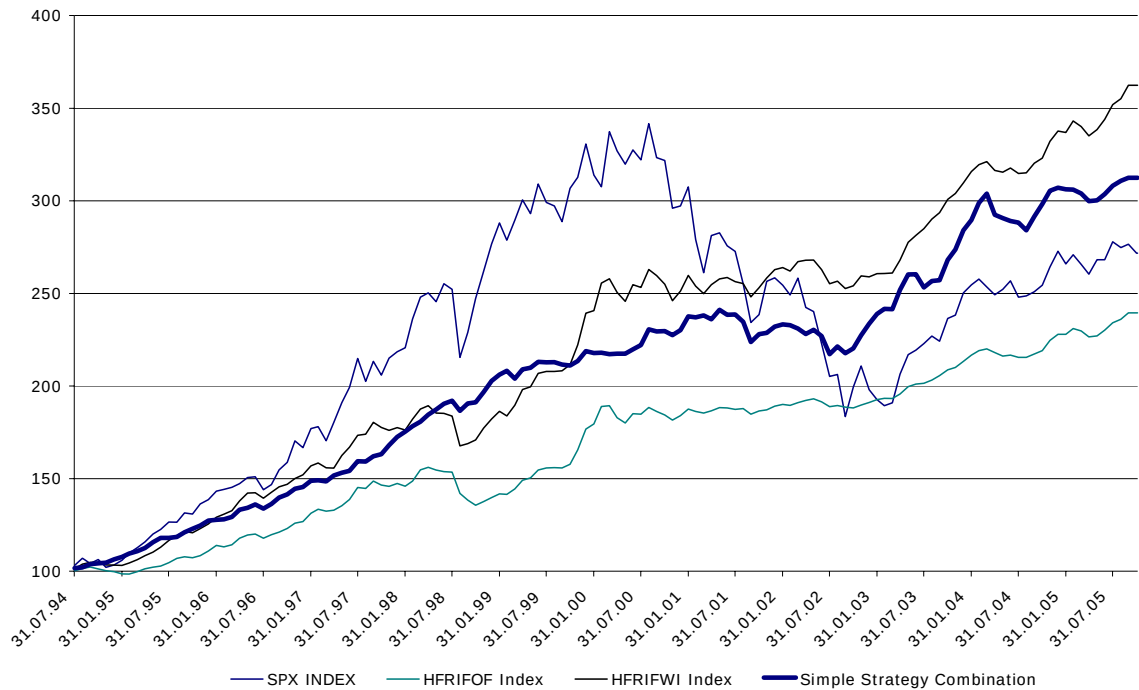


Fig. 2: Performance of an equally weighed combination of three strategies: the sgfi trend following index, the BXM covered call writing index, and long the Credit Suisse High Yield Bond Index (annualised return: 10.3%, annualised volatility: 5.6%). For comparison, we show the performance of the HFR Composite (annualised return: 11.7%, annualised volatility: 7.2%), the HFR Fund of Funds Index (annualised return: 7.9%, annualised volatility: 5.8%) and the S&P 500.

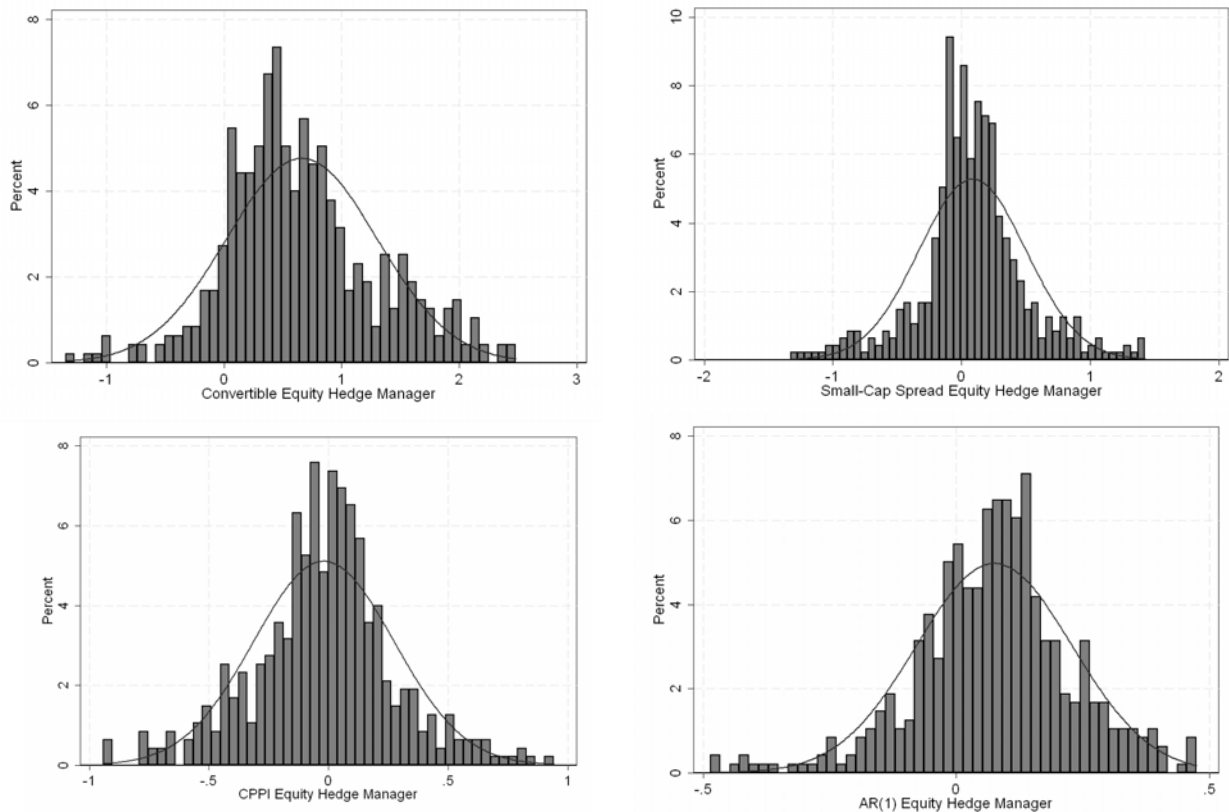


Fig. 3 Histogram of the factor exposures ("betas") of Long/Short Equity managers using the independent variable as in Table 1. Data: Tass

Alternative Factors

HFR Index 01/94 - 12/04	Asset Class Factor	Beta (Alpha)	t-value (absolute)	Adj. R2
Equity Hedge (Long/Short Equity)	Citigroup Convertible	0.545	19.49	88.5%
	Small-Cap Spread (Wilshire)	0.179	6.42	
	CPPI S&P 12M	0.181	4.31	
	AR(1)	0.125	4.05	
	Alpha	0.350	3.80	
Equity Market Neutral	Fama-French UMD	0.101	7.94	35.3%
	S&P 500	0.070	4.63	
	Value Spread (MSCI)	0.063	2.59	
	Small-Cap Spread (Wilshire)	0.035	1.90	
	Alpha	0.215	3.47	
Short Selling	Citigroup Convertible	-1.042	-8.78	81.2%
	Value Spread (MSCI)	0.554	5.93	
	S&P 600 Small Cap	-0.303	-3.97	
	Alpha	0.195	0.80	
Event Driven	S&P 500	0.254	13.41	79.3%
	Small-Cap Spread (Wilshire)	0.233	9.81	
	CSFB High Yield	0.255	5.03	
	AR(1)	0.155	3.64	
	Alpha	0.432	5.37	
Distressed	AR(1)	0.386	7.38	68.4%
	S&P 500	0.151	7.33	
	Small-Cap Spread (Wilshire)	0.145	5.71	
	CSFB High Yield	0.280	5.21	
	Alpha	0.240	2.79	
Merger Arbitrage	S&P 600 Small Cap	0.071	4.22	52.9%
	Russell 3000 Value	0.057	3.52	
	BXM Covered Call Writing Index	0.092	3.06	
	Merger Fund	0.077	2.90	
	Alpha	0.328	4.72	
Fixed Income Arbitrage	Citigroup Convertible	0.276	5.59	40.5%
	JP Morgan EM Global Bond	0.104	3.04	
	AR(1)	0.140	2.01	
	Credit Spread (BB vs AAA)	0.026	3.10	
	Alpha	0.227	1.50	
Convertible Arb.	AR(1)	0.427	7.07	54.0%
	Citigroup Conv Inv. Grade	0.242	5.03	
	CSFB High Yield	0.173	4.71	
	S&P 500	-0.085	-3.08	
	Alpha	0.160	2.54	
Macro	Lehman World Gov. Bond	0.518	4.23	49.7%
	Citigroup Convertible	0.231	4.21	
	sGFI	0.164	4.08	
	MSCI EM Global Equity	0.086	3.02	
	Alpha	0.196	1.43	
Managed Futures	sGFI	0.343	6.55	34.3%
	Lehman World Gov. Bond	0.442	2.75	
	Goldman Commodity Index	0.075	2.37	
	Alpha	-0.027	-0.15	
Managed Futures Trend Followers	sGFI	0.584	6.85	35.4%
	Lehman World Gov. Bond	0.709	2.71	
	Goldman Commodity Index	0.110	2.12	
	Alpha	-0.156	-0.53	

Table 1: Results of linear asset class factor modelling for the different hedge fund strategies with a broader set of risk factors (based on monthly data: HFR; for Managed Futures: CISDM Managed Futures Qualified Universe and Trend Following Indices, data from Jan. 94 to Dec 2004).

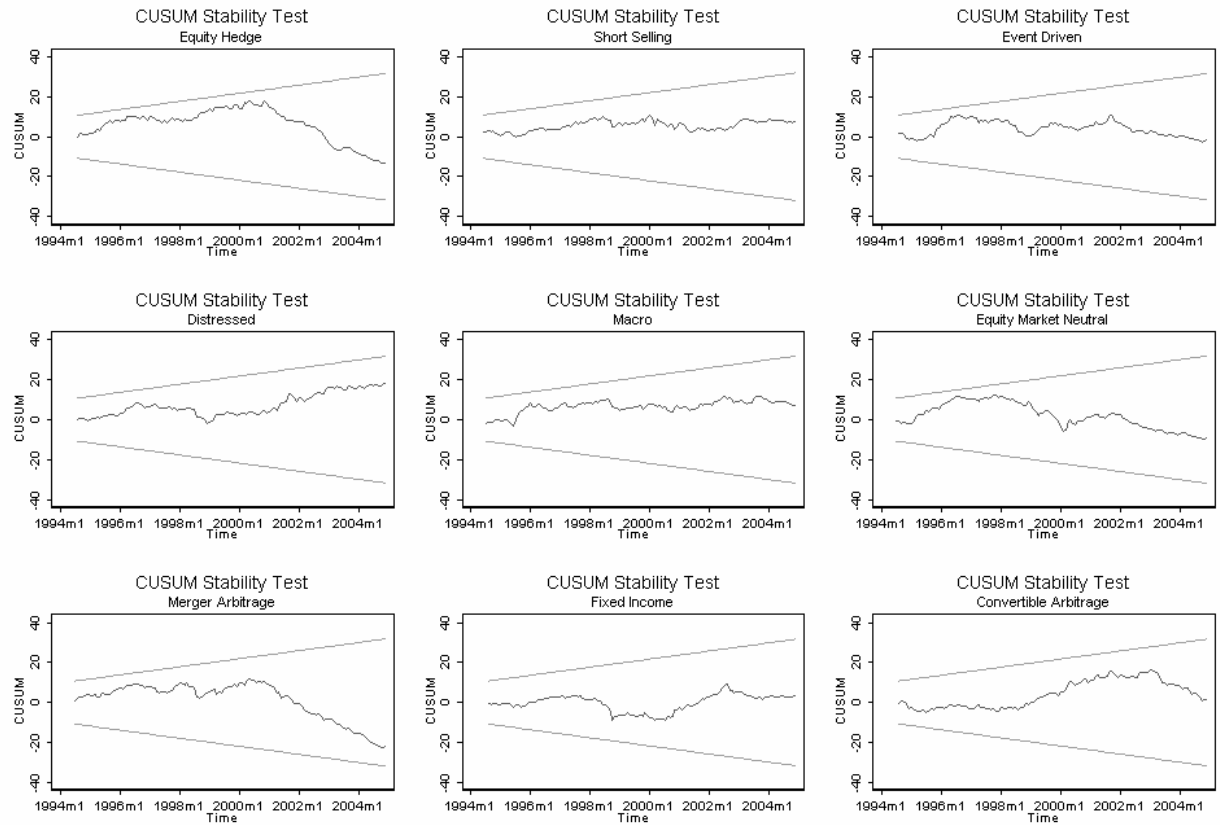
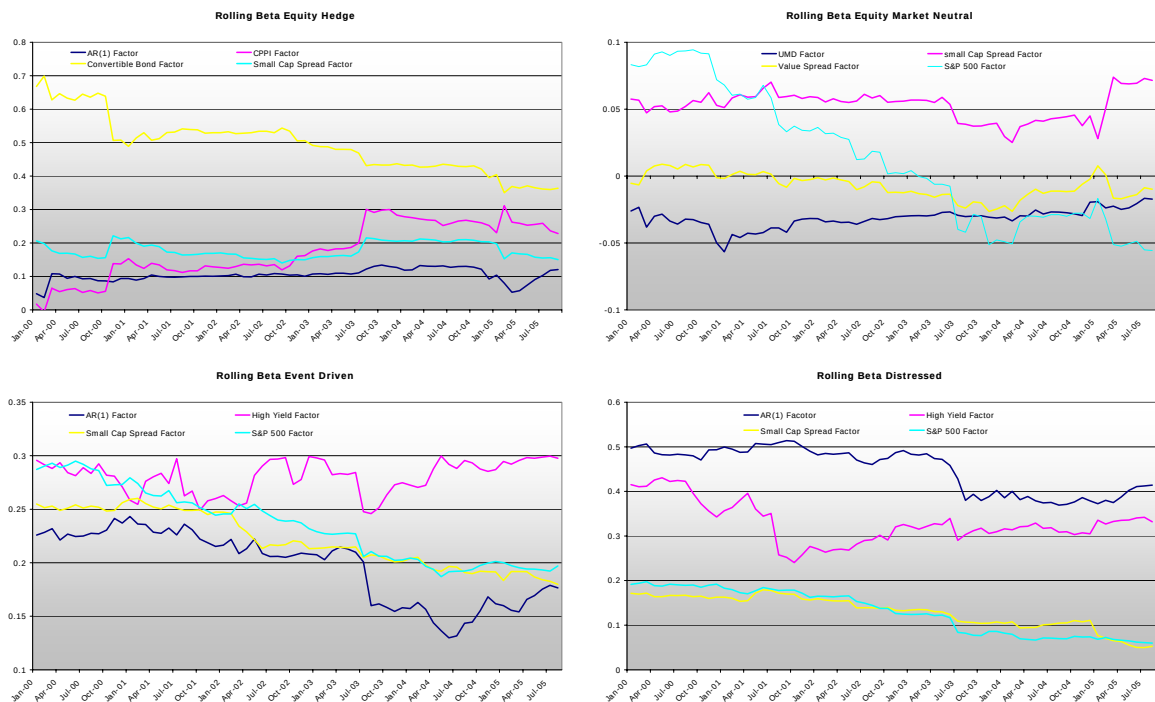


Fig. 4: Results of a CUSUM stability test for the regression models in table 1



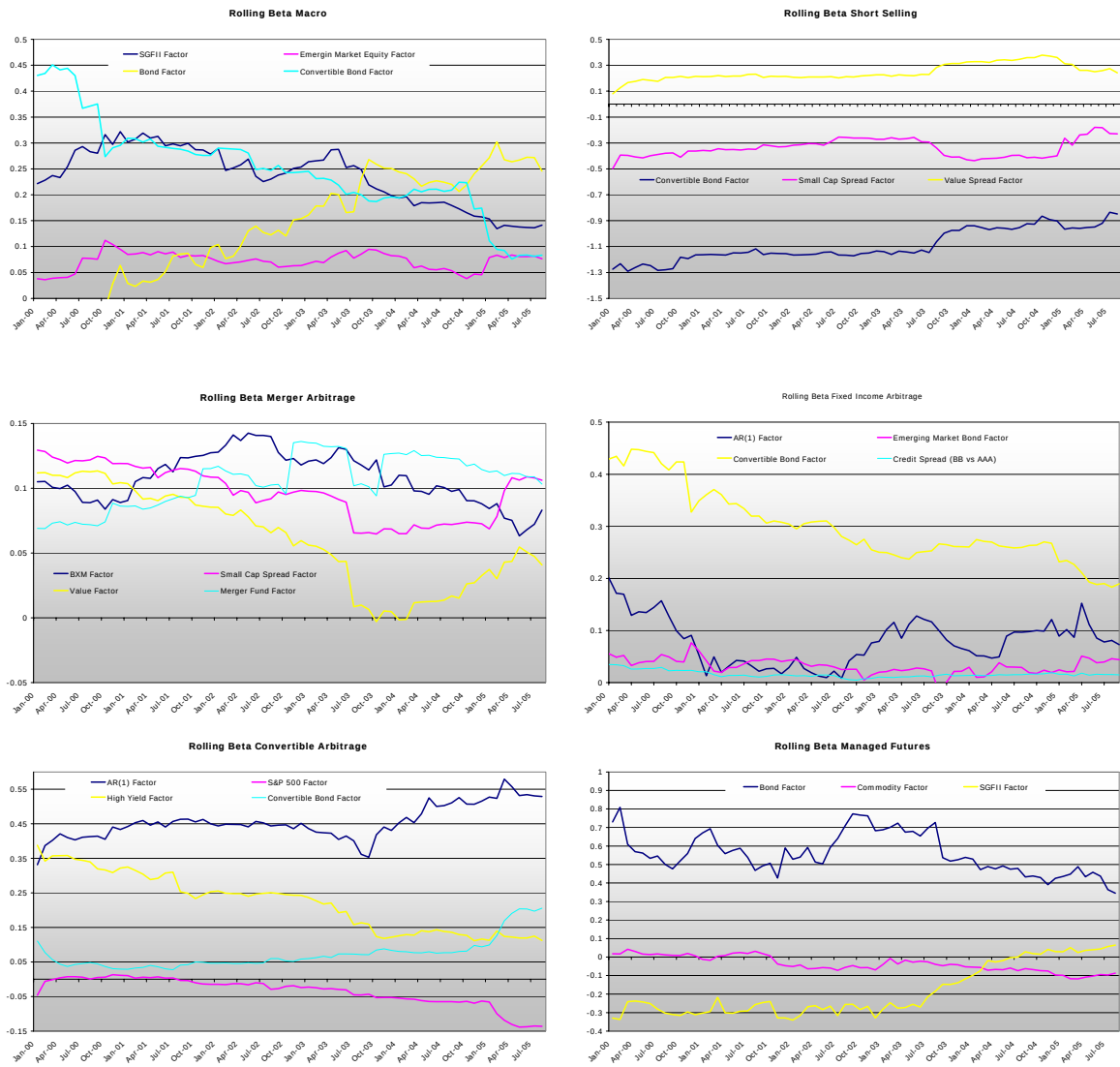
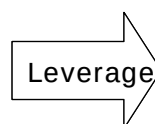
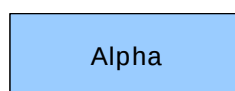
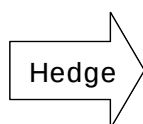


Fig. 5: Factor exposures for the regression models in Table 1 as they developed employing a rolling regression with a 60-month time window

Active long-only bonds/equity fund



Hedge Fund



Fig. 6: A schematic model for hedge fund return sources based on results in Table 1

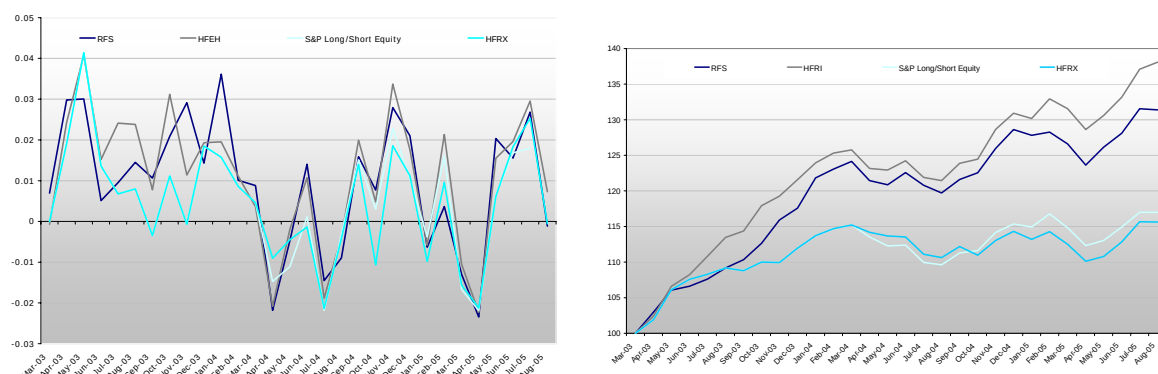


Fig. 7: Returns (monthly and cumulated) of the non-investable HFRI Equity Hedge Index, the investable HFRX Equity Hedge Index, and the (investible) S&P Long/Short Equity Index (all in light color) versus the RFS cumulative return (in dark color) based on the factor returns (see text for details).

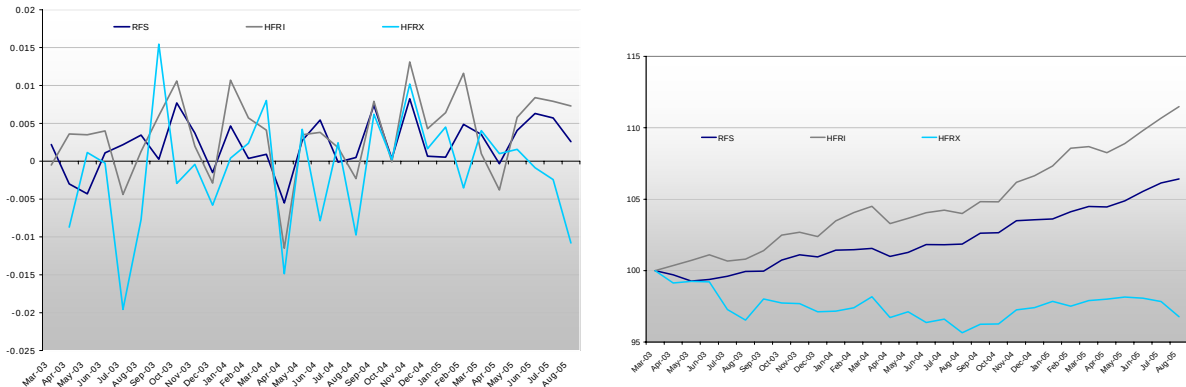


Fig. 8: Returns (monthly and cumulated) of the non-investable HFRI Equity Market Neutral Index and the investable HFRX Equity Market Neutral Index (in light color) versus the RFS cumulative return (in dark color) based on the factor returns (see text for details).

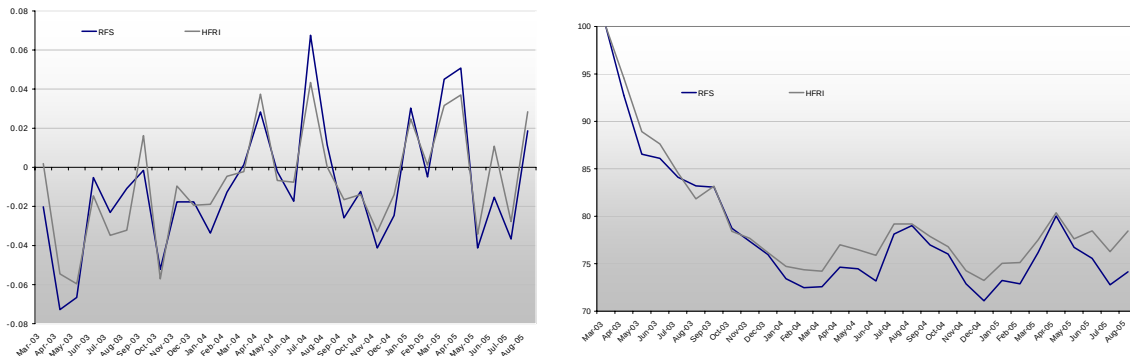


Fig. 9: Returns (monthly and cumulated) of the non-investable HFRI Dedicated Short Bias Index versus the RFS cumulative return (in dark color) based on the factor returns (see text for details). Note: An investable version of the HFR index does not exist for dedicated short hedge funds

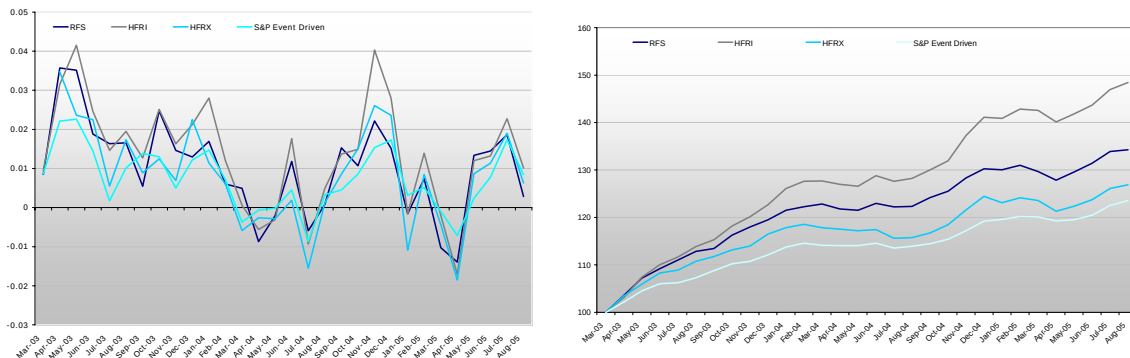


Fig. 10: Returns (monthly and cumulated) of the non-investable HFRI Event Driven Index and the investable HFRX Event Driven Index (in light color) versus the RFS cumulative return (in dark color) based on the factor returns (see text for details).

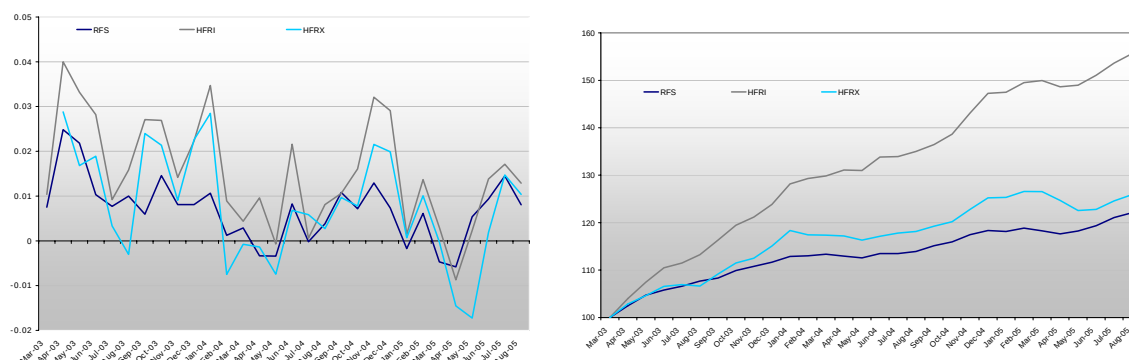


Fig. 11: Returns (monthly and cumulated) of the non-investable HFRI Distressed Index and the investable HFRX Distressed Index (in light color) versus the RFS cumulative return (in dark color) based on the factor returns (see text for details).

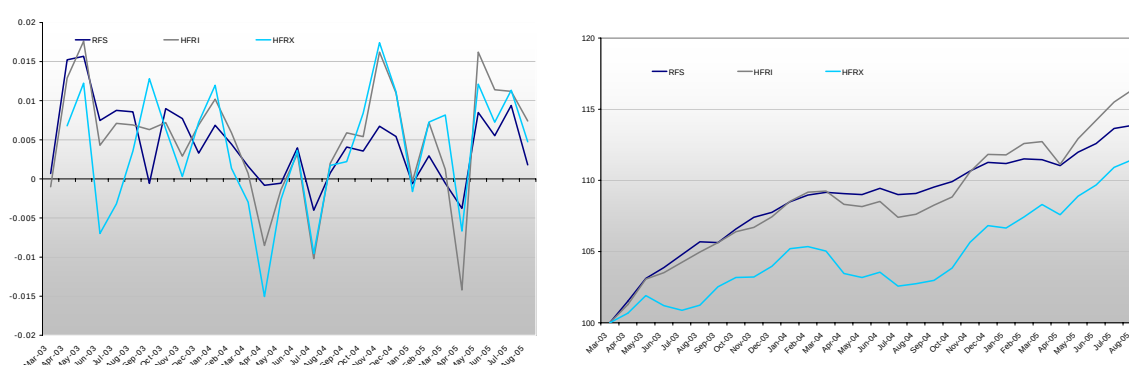


Fig. 12: Returns (monthly and cumulated) of the non-investable HFRI Merger Arbitrage Index and the investable HFRX Merger Arbitrage Index (in light color) versus the RFS cumulative return (in dark color) based on the factor returns (see text for details).

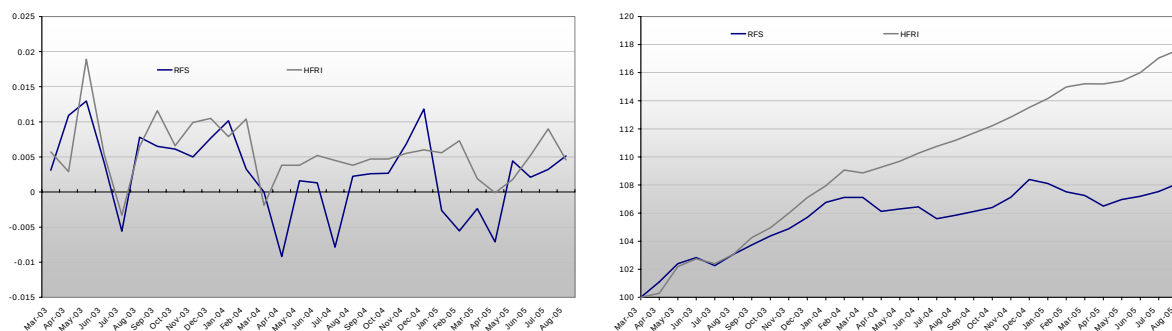


Fig. 13: Returns (monthly and cumulated) of the non-investable HFRI Fixed Income Index versus the RFS cumulative return (in dark color) based on the factor returns (see text for details). Note: An investable version of the HFR index does not exist for Fixed Income Arbitrage hedge funds.

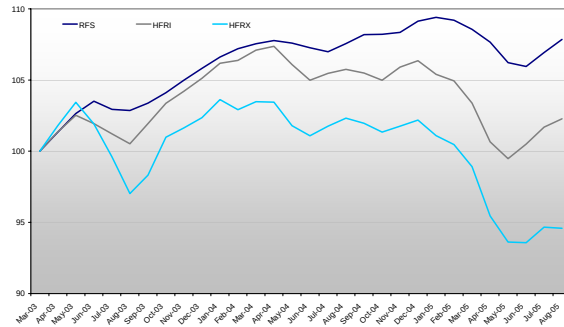
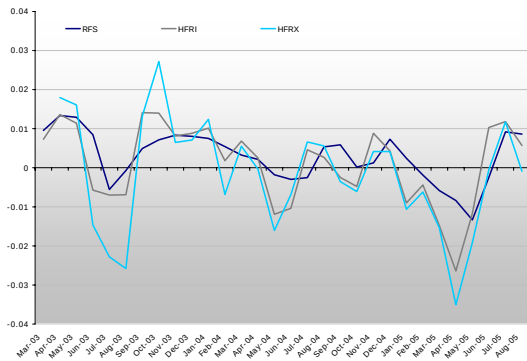


Fig. 14: Returns (monthly and cumulated) of the non-investable HFRI Convertible Arbitrage Index and the investable HFRX Convertible Arbitrage Index (in light color) versus the RFS cumulative return (in dark color) based on the factor returns (see text for details).

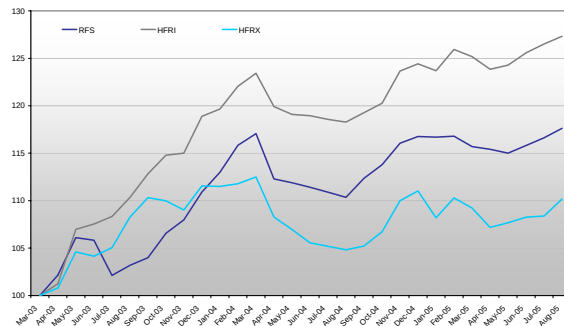
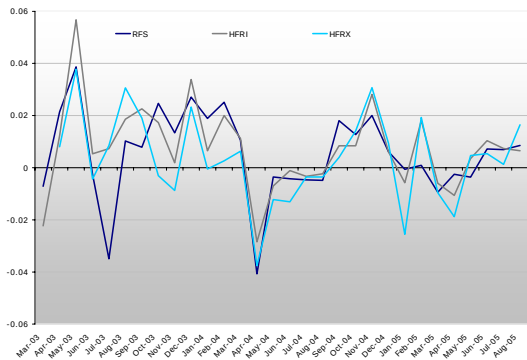


Fig. 15: Returns (monthly and cumulated) of the non-investable HFRI Global Macro Index and the investable HFRX Global Macro Index (in light color) versus the RFS cumulative return (in dark color) based on the factor returns (see text for details).

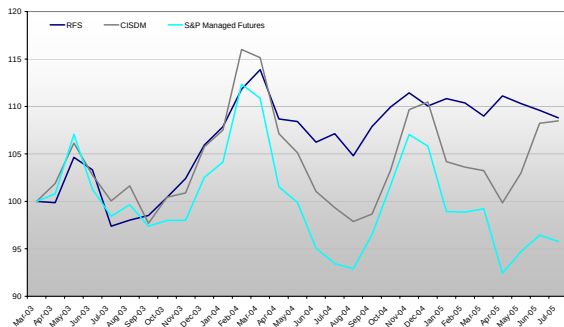
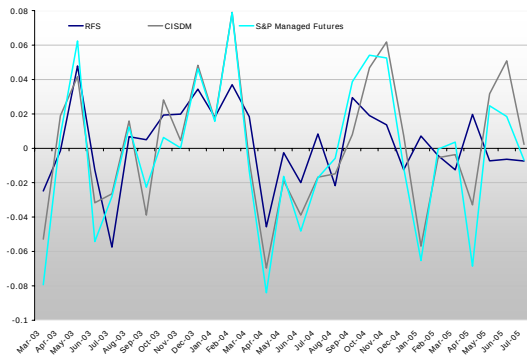


Fig. 16: Returns (monthly and cumulated) of the non-investable (!) CISDM Managed Futures Qualified Universe Index (in grey color) versus the RFS cumulative return (in dark color) based on the factor returns (see text for details).

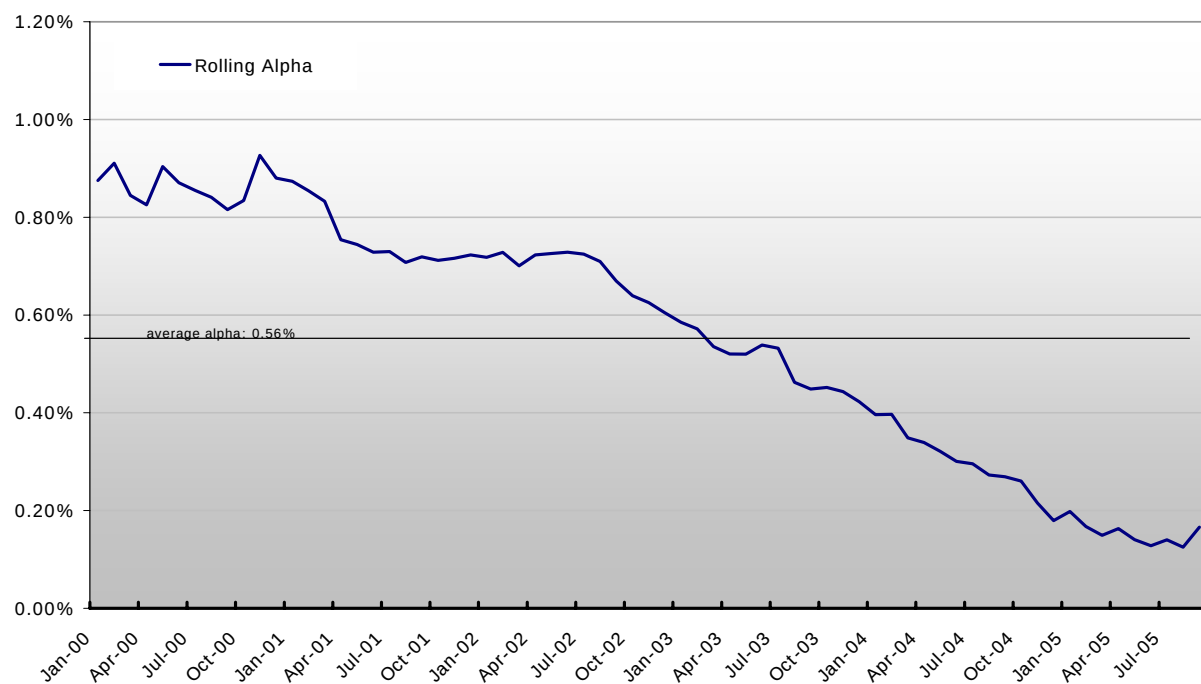


Fig. 17: The development of alpha for Long/Short Equity funds (HFR sub-index) based on a rolling regression over a 60 month time window. The risk factors were chosen as in Table 1.

Strategy	RFS	HFRX	HFR I
Equity Hedge	27.8%	16.0%	32.8%
Market Neutral	6.2%	-3.9%	10.9%
Short Selling	-28.2%	N/A	-23.0%
Event Driven	29.8%	24.1%	40.0%
Distressed	20.1%	23.3%	44.8%
Merger Arbitrage	13.0%	10.9%	15.3%
Fixed Income	7.8%	N/A	16.3%
Convertible Arbitrage	7.6%	-5.3%	2.4%
Global Macro	16.7%	10.1%	24.6%
Managed Futures	9.2%	N/A	N/A

Table 2: Cumulated performance of the RFS and the HFRX strategy, data from March 2003 to August 2005.